

LTE Carrier Aggregation Technology Development and Deployment Worldwide



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1. EXECUTIVE SUMMARY

If there are any sure bets about the future of mobile, they are that data usage will continue skyrocketing, new applications and services will require even higher speeds and the global base of consumer and business customers will continue to grow. Today's development work on flexible and forward-looking technologies such as Carrier Aggregation (CA) will enable operators, vendors, content providers and the rest of the mobile ecosystem to meet this growing demand.

CA is a key feature of LTE-Advanced that enables operators to create larger "virtual" carrier bandwidths for LTE services by combining separate spectrum allocations. As of August 2014, CA is the primary feature deployed by operators with commercial LTE-Advanced service. The need for CA in LTE-Advanced arises from the requirement to support bandwidths larger than those currently supported in LTE (up to 20 MHz) while at the same time ensuring backward compatibility with LTE. The benefits of this aggregation include higher peak data rates and increased average data rates for users.

CA enables the combination of up to five LTE Release 8 (Rel-8) compatible carriers. As a result, operators can provide high throughput without wide contiguous frequency band allocations, and they can ensure statistical multiplexing gain by distributing the traffic dynamically over multiple carriers. With CA, operators also can take asymmetrical bands into FDD use because there can be downlink-only frequency bands.

CA combinations are specified within 3GPP RAN Working Group 4. CA combinations are divided into intra-band (contiguous and non-contiguous) and inter-band. Intra-band contiguous and inter-band combinations, aggregating two Component Carriers (CCs) in downlink, are specified in 3GPP Rel-10. 3GPP Rel-11 offers many more CA configurations, including non-contiguous intra-band CA and Band 29 for inter-band CA, which is also referred as Supplemental Downlink. Release 12 will include CA of FDD and TDD frequency bands, as well as support of aggregation of two CCs in uplink and three CCs in downlink.

CA can currently provide bandwidths up to 100 MHz. Aggregated carriers can be adjacent or non-adjacent even at different frequency bands. The result is that basically all of an operator's frequency allocations can be used to provide LTE services. CA provides spectrum efficiency and peak rates nearly on par with single wideband allocation. In some heterogeneous deployment scenarios, CA performance can be even better because flexible frequency reuse can be arranged between local area nodes to provide better inter-cell interference coordination.

CA supports cross component carrier scheduling, where the control channel at one carrier can be used to allocate resources at another carrier for user data transmission. It can be used to provide both frequency diversity and interference coordination in frequency domain at the same time, underlining its significance as a powerful technology for effective utilization of radio resources. Note that the uplink gain tends to be lower than the downlink gain because the User Equipment (UE) cannot always utilize multi-carrier transmission due to limited transmit power.

If carriers are at different frequency bands, they have different propagation losses and different interfering systems, all of which affect achievable data rates, transmit power and usage of resources. For example, a UE on the fringe of a cell site's coverage could be better served with a low-frequency carrier, while a UE a high-frequency carrier is a better fit for a UE close to the site. Inter-band CA provides more flexibility to utilize fragmented spectrum allocations.

CA has also been designed to be a future-proof technology, with great potential into and beyond Release 12. By extending aggregation to more carriers and enabling aggregation of additional licensed spectrum, CA will play a key role in enabling both IMT-Advanced and the use of emerging spectrum allocations. There also will be multiple and varied deployments of CA tailored to operators' specific requirements.

However, this seemingly simple concept will continue to stretch the realm of network and device implementations. Strong collaboration across the industry for further innovations, evolutions and optimization will drive CA as a mainstream technology. In particular, operators are advised to carry out consultations with their RAN infrastructure vendors, device suppliers and chipset manufacturers when developing their CA implementation plans.

2. INTRODUCTION [1]

The first version of LTE was completed in March 2009 as part of Third Generation Partnership Program (3GPP) Rel-8. LTE is based on a flat radio access network architecture without a centralized network component. LTE offers flexible bandwidth options ranging from 1.4 to 20 MHz using orthogonal frequency-division multiple access (OFDMA) in the downlink and single-carrier, frequency-division multiple access (SC-FDMA) in the uplink.

In March 2008, 3GPP started a new study item to further develop LTE toward LTE-Advanced by targeting the IMT-Advanced requirements as defined by the International Telecommunication Union (ITU). The LTE-Advanced study item was closed in March 2010. The outcome was a set of new radio features, which are currently being standardized to become part of LTE-Advanced. CA is one of the main features of LTE-Advanced in Rel-10 for meeting the peak data rate requirements of IMT-Advanced: 3 Gbps DL and 1.5 Gbps UL¹. With CA, users gain access to a total bandwidth of up to 100 MHz in order to meet the IMT-Advanced requirements. The system bandwidth may be contiguous, or composed of several non-contiguous bandwidth chunks that are aggregated. CA is designed to be backward compatible, meaning that legacy Rel-8 and Rel-9 users should be able to coexist with LTE-Advanced on at least part of the total bandwidth.

This white paper presents a summary of CA technology ranging from the basic concepts to future evolutions. It also provides an overview of supported CA configurations and combinations specified in 3GPP, CA control mechanisms, common CA deployment scenarios, specific use cases, device implementations and performance impacts. The paper also summarizes future enhancements that will continue to drive the evolution of CA technology and deployment as a mainstream technology. A glossary section at the end defines the acronyms and terms used throughout this paper.

3. GLOBAL MARKET TRENDS, MILESTONES IN CARRIER AGGREGATION (CA)

Mobile operators continue to seek efficient and cost-effective solutions to support the high demand for data and next-generation services. Spectrum continues to be a limiting factor. A majority of operators have spectrum allocations of less than 20 MHz contiguous spectrum or have multiple bands or non-contiguous spectrum blocks. As a solution, operators worldwide are adopting CA technology, which is one of the main features of LTE-Advanced in Rel-10. Over the long term, CA will enable operators to meet the peak requirements of IMT-Advanced: 3 Gbps DL and 1.5 Gbps UL.

¹ See <http://www.3gpp.org/technologies/keywords-acronyms/97-lte-advanced>

Although it is a relatively new technology, CA already has significant marketplace traction. Some examples:

- The first commercial CA deployments were in 2013 in South Korea, where SK Telecom aggregated 20 MHz of 800MHz and 1.8GHz spectrum to achieve peak DL rates of 150Mbps. LG UPlus followed a month later [17].
- In November 2013, EE in the U.K. launched 40 MHz of inter-band CA with 300 Mbps theoretical speeds [18].
- Australian mobile operator Optus was the first to deploy CA on a TD-LTE network in December 2013 [19]. Several other operators held trials or announced plans to deploy CA such as A1 Telekom in Austria, Softbank Japan, CSL Limited Hong Kong and Telstra Australia. [27]

In North America, AT&T announced CA capability in Chicago and plans for other markets in aggregating 2100 MHz and 700 MHz spectrum to deliver 15 MHz of DL spectrum [20]. Sprint plans to use CA as part of the Sprint Spark service to combine LTE spectrum across the 800 MHz, 1900 MHz and 2.5 GHz spectrum. T-Mobile & Verizon plan to use carrier aggregation to further enhance their AWS spectrum holdings.

- Apple, Samsung, LG and HTC have launched CA-capable devices such as the Samsung Galaxy S 4, Note 3, LG Pro2, the HTC One M8 and Apple iPhone 6 and 6 Plus [17], [21], [22], [26].

Initial CA deployments on network and devices have been mostly limited to two carriers. However, there are plans to deploy three carriers starting in 2015. CA with three carriers was successfully demonstrated early 2014 by operators like SK Telecom, LG UPlus and Telstra [23], [24], [25].

CA also has great potential for future developments such as increased number of component carriers, bands and bandwidths, enabling aggregation of FDD, TDD, licensed and unlicensed spectrum. In September 2014, China Telecom announced the successful completion of an FDD-TDD CA demonstration. [28]

4. 3GPP STATUS FOR RELEASE 10, RELEASE 11 AND RELEASE 12 CA FEATURES

4.1 3GPP CA COMBINATIONS DEFINITION PROCESS

CA combinations are specified within 3GPP RAN Working Group 4. UE requirements are defined in Technical Specification (TS) 36.101, while Base Station (BS) requirements are specified in TS 36.104. Before UE and BS requirements are specified in relevant specifications, the CA Work Item (WI) needs to be opened in 3GPP RAN. This requires submission of a Work Item Description (WID), which contains details related to proposed CA combination to be specified. This information includes which frequency bands will be aggregated, the channel bandwidths considered for this combination and the timeline to define this in 3GPP. In order to open WI in 3GPP, the support of at least four 3GPP members is required. Once a new WI is opened, technical work takes place in 3GPP RAN4, including harmonic and intermodulation products analysis, as well as the impact on UE and BS requirements. After the consensus is reached, Change Requests (CRs) to relevant specification are proposed for approval. With

agreement in 3GPP RAN4, CRs are formally approved in 3GPP RAN, WI is closed and new versions of specifications are created.

Table 1 describes LTE frequency bands defined in 3GPP [2] that will be referred to throughout this paper.

Table 1. LTE Operating Bands.

LTE Operating Band	Uplink (UL) operating band BS receive UE transmit		Downlink (DL) operating band BS transmit UE receive		Duplex Mode
	F_{UL_low}	F_{UL_high}	F_{DL_low}	F_{DL_high}	
1	1920 MHz	– 1980 MHz	2110 MHz	– 2170 MHz	FDD
2	1850 MHz	– 1910 MHz	1930 MHz	– 1990 MHz	FDD
3	1710 MHz	– 1785 MHz	1805 MHz	– 1880 MHz	FDD
4	1710 MHz	– 1755 MHz	2110 MHz	– 2155 MHz	FDD
5	824 MHz	– 849 MHz	869 MHz	– 894 MHz	FDD
6 ¹	830 MHz	– 840 MHz	875 MHz	– 885 MHz	FDD
7	2500 MHz	– 2570 MHz	2620 MHz	– 2690 MHz	FDD
8	880 MHz	– 915 MHz	925 MHz	– 960 MHz	FDD
9	1749.9 MHz	– 1784.9 MHz	1844.9 MHz	– 1879.9 MHz	FDD
10	1710 MHz	– 1770 MHz	2110 MHz	– 2170 MHz	FDD
11	1427.9 MHz	– 1447.9 MHz	1475.9 MHz	– 1495.9 MHz	FDD
12	699 MHz	– 716 MHz	729 MHz	– 746 MHz	FDD
13	777 MHz	– 787 MHz	746 MHz	– 756 MHz	FDD
14	788 MHz	– 798 MHz	758 MHz	– 768 MHz	FDD
15	Reserved		Reserved		FDD
16	Reserved		Reserved		FDD
17	704 MHz	– 716 MHz	734 MHz	– 746 MHz	FDD
18	815 MHz	– 830 MHz	860 MHz	– 875 MHz	FDD
19	830 MHz	– 845 MHz	875 MHz	– 890 MHz	FDD

20	832 MHz	–	862 MHz	791 MHz	–	821 MHz	FDD
21	1447.9 MHz	–	1462.9 MHz	1495.9 MHz	–	1510.9 MHz	FDD
22	3410 MHz	–	3490 MHz	3510 MHz	–	3590 MHz	FDD
23	2000 MHz	–	2020 MHz	2180 MHz	–	2200 MHz	FDD
24	1626.5 MHz	–	1660.5 MHz	1525 MHz	–	1559 MHz	FDD
25	1850 MHz	–	1915 MHz	1930 MHz	–	1995 MHz	FDD
26	814 MHz	–	849 MHz	859 MHz	–	894 MHz	FDD
27	807 MHz	–	824 MHz	852 MHz	–	869 MHz	FDD
28	703 MHz	–	748 MHz	758 MHz	–	803 MHz	FDD
29	N/A			717 MHz	–	728 MHz	FDD ²
30	2305 MHz	–	2315 MHz	2350 MHz	–	2360 MHz	FDD
31	452.5 MHz	–	457.5 MHz	462.5 MHz	–	467.5 MHz	FDD
32	N/A			1452 MHz	–	1496 MHz	FDD ²
33	1900 MHz	–	1920 MHz	1900 MHz	–	1920 MHz	TDD
34	2010 MHz	–	2025 MHz	2010 MHz	–	2025 MHz	TDD
35	1850 MHz	–	1910 MHz	1850 MHz	–	1910 MHz	TDD
36	1930 MHz	–	1990 MHz	1930 MHz	–	1990 MHz	TDD
37	1910 MHz	–	1930 MHz	1910 MHz	–	1930 MHz	TDD
38	2570 MHz	–	2620 MHz	2570 MHz	–	2620 MHz	TDD
39	1880 MHz	–	1920 MHz	1880 MHz	–	1920 MHz	TDD
40	2300 MHz	–	2400 MHz	2300 MHz	–	2400 MHz	TDD

41	2496 MHz	2690 MHz	2496 MHz	2690 MHz	TDD
42	3400 MHz	– 3600 MHz	3400 MHz	– 3600 MHz	TDD
43	3600 MHz	– 3800 MHz	3600 MHz	– 3800 MHz	TDD
44	703 MHz	– 803 MHz	703 MHz	– 803 MHz	TDD
NOTE 1: Band 6 is not applicable					
NOTE 2: Restricted to E-UTRA operation when Carrier Aggregation is configured. The downlink operating band is paired with the uplink operating band (external) of the Carrier Aggregation configuration that is supporting the configured Pcell.					

4.2 CA CONFIGURATIONS

CA combinations are divided into intra-band (contiguous and non-contiguous) and inter-band. Intra-band contiguous and inter-band combinations, aggregating two Component Carriers (CCs) in downlink, are specified from Release 10. The Intra-band contiguous CA configuration refers to contiguous carriers aggregated in the same operating band. The Intra-band non-contiguous CA configuration refers to non-contiguous carriers aggregated in the same operating band. The Inter-band CA configuration refers to aggregation of component carriers in different operating bands, where the carriers aggregated in each band can be contiguous or non-contiguous. The spacing between the center frequencies of two contiguous CCs is $N \times 300$ kHz, where N =integer, and for non-contiguous cases, the CCs are separated by one, or more, frequency gap(s).

3GPP Rel-10 is the first release to support CA. It defines a limited variety of CA configurations, including: contiguous intra-band CA for bands 1 (FDD) and band 40 (TDD) named CA_1C and CA_40C respectively and an inter-band CA between bands 1 and 5, named CA_1A-5A.

3GPP Rel-11 offers many more CA configurations, including non-contiguous intra-band CA and Band 29 for inter-band CA, which is also referred as supplemental DL. Band 29 is a special FDD band in Rel-11 that has only a DL component and no UL component. The intention is that this band is available for bandwidth expansion only and holds the functionality of DL Secondary cell (SCell). This serving cell is referred to as the primary cell (PCell). In the downlink, the carrier corresponding to the PCell is the Downlink Primary Component Carrier (DL PCC), while in the uplink it is the Uplink Primary Component Carrier (UL PCC). Other serving cells are referred to as secondary cells (SCells) and are used for bandwidth expansion for the particular UE. It is to be noted that there are additional differences in the functionality. More details about SCell can be found in Section 5.2.

Release 12 will include CA of FDD and TDD frequency bands, as well as support for aggregating two UL CCs and three DL CCs. Note that even though certain CA combinations are specified in later releases, these are release independent and can be supported from previous releases' equipment due to backward-compatible signaling. 3GPP Rel-12 offers aggregation of three DL CCs, support for CA configurations for UL inter-band and intra-band non-contiguous CA and band 32 support (1.5GHz L Band), which has a DL component only and further enhanced capabilities that are still under study.

Figure 1a presents the 3GPP Release Status for Carrier Aggregation as of September 2014. Figure 1b presents the CA bands in Re-10 and Rel-11 and Figure 1c presents the CA bands in Rel-12 as of September 2014.

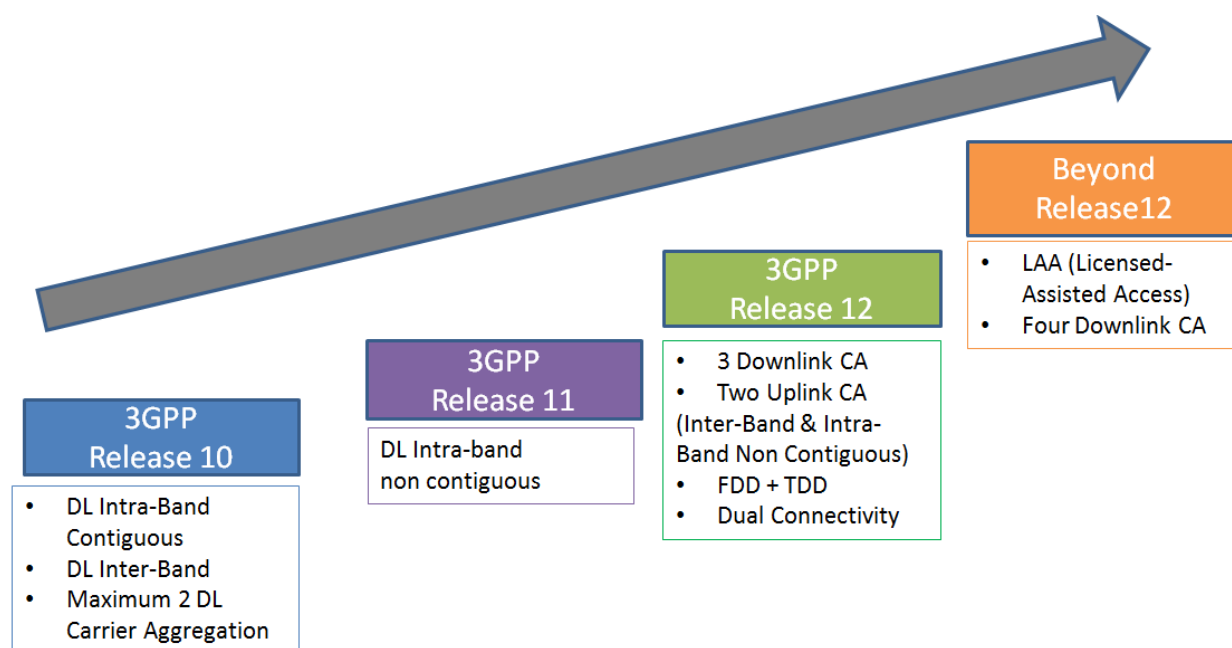


Figure 1a. 3GPP Release Status for Carrier Aggregation.

Carrier aggregation bands in 3GPP Rel-10
(Source: TS36.104)

Intra-band contiguous CA (2DL/2UL):

CA Band	E-UTRA operating band
CA C 1	1
CA C 40	40

Inter-band CA (2DL/1UL):

CA Band	E-UTRA operating bands
CA_1-5	1
	5

Carrier Aggregation bands in 3GPP Rel-11

Inter-band CA (2DL/1UL):

CA Band	E-UTRA operating band	Requested by
CA_1-19	1+19	NTT DOCOMO
CA_3-7	3+7	TeliaSonera
CA_4-13	4+13	Verizon Wireless
CA_4-17	4+17	AT&T
CA_7-20	7+20	Orange et al
CA_5-12	5+12	US Cellular
CA_4-12	4+12	Cox Communication
CA_2-17	2+17	AT&T
CA_4-5	4+5	AT&T
CA_5-17	5+17	AT&T
CA_3-5	3+5	SK Telecom
CA_4-7	4+7	Rogers Wireless
CA_3-20	3+20	Vodafone
CA_8-20	8+20	Vodafone
CA_1-18	1+18	KDDI
CA_1-21	1+21	NTT DOCOMO
CA_11-18	11+18	KDDI
CA_3-8	3+8	KT
CA_2-29	2+29	AT&T
CA_4-29	4+29	AT&T

Intra-band contiguous and DL non-contiguous CA:

CA Band	E-UTRA operating band	Requested by
CA_C_41	41 (2DL/2UL)	Clearwire, CMCC,...
CA_C_38	38 (2DL/2UL)	CMCC
CA_C_7	7 (2DL/2UL)	CUC, CT, Telenor et al
CA_NC_B25	25 (2DL/1UL)	Sprint
CA_NC_B41	41 (2DL/1UL)	CMCC

Figure 1b. 3GPP Carrier Aggregation Bands in Rel-10 and Rel-11.

Carrier Aggregation bands in 3GPP Rel-12

Inter-band CA (2DL/1UL):

CA Band	E-UTRA operating band	Requested by
CA 1-3	1 + 3	China Unicom, China Telecom
CA 1-7	1 + 7	LGU +
CA 1-8	1 + 8	Softbank
CA 1-11	1 + 11	Softbank
CA 1-18	1 + 18	KDDI
CA 1-26	1 + 26	KDDI
CA 2-4	2 + 4	TMO-US
CA 2-5	2 + 5	AT&T
CA 2-12	2 + 12	US Cellular
CA 2-13	2 + 13	Verizon
CA 3-19	3 + 19	NTT DOCOMO
CA 3-20	3 + 20	Telekom Austria
CA 3-26	3 + 26	KT
CA 3-27	3 + 27	KT
CA 3-28	3 + 28	eAccess
CA 4-12	4 + 12	TMO-US
CA 4-27	4 + 27	NII Holdings
CA 5-7	5 + 7	LG Uplus
CA 5-25	5 + 25	US Cellular
CA 7-20	7 + 20	Telekom Austria
CA 7-28	7 + 28	Telefonica
CA 8-11	8 + 11	Softbank
CA 8-20	8 + 20	Vodafone
CA 12-25	12 + 25	US Cellular
CA 19-21	19 + 21	NTT DOCOMO
CA 20-32	20 + 32 (in UTRA B1 + BXXXII)	Orange
CA 23-29	23 + 29	Dish
CA 39-41	39 + 41	CMCC
CA 41-42	41 + 42	China Unicom, China Telecom

Carrier Aggregation bands in 3GPP Rel-12

Intra-band contiguous CA:

CA Band	E-UTRA operating band	Requested by
CA C B3	3 (2DL/2UL)	Chin Unicom
CA C B7	7 (2DL/2UL)	Orange
CA C B23	23 (2DL/1UL)	Dish
CA C B27	27 (2DL/1UL)	NII Holdings
CA C B39	39 (2DL/2UL)	CMCC
CA C B40	40 (3DL/1UL)	CMCC
CA C B42	42 (2DL/2UL)	CMCC, NII, Boflore

Intra-band DL non-contiguous CA:

CA Band	E-UTRA operating band	Requested by
CA NC B2	2 (2DL/1UL)	Verizon
CA NC B3	3 (2DL/1UL)	SK Telecom
CA NC B4	4 (2DL/1UL)	TMO-US
CA NC B7	7 (2DL/1UL)	Telecom Italia
CA NC B23	23 (2DL/1UL)	Dish
CA NC B25	25 (2DL/1UL)	Telus
CA NC B42	42 (2DL/1UL)	CMCC, NII, Boflore

Inter-band TDD-FDD CA (2DL/1UL):

CA Band	E-UTRA operating band	Requested by
CA 8-40	8 + 40	KT
CA 1-42	1 + 42	NTT DOCOMO
CA 19-42	19 + 42	NTT DOCOMO

Carrier Aggregation bands in 3GPP Rel-12

Inter-band 3DL CA (including intra-band non-contiguous CA):

CA Band	E-UTRA operating band	Requested by
CA 1-3-5	1 + 3 + 5	SK Telecom
CA 1-3-8	1 + 3 + 8	KT
CA 1-3-19	1 + 3 + 19	NTT DOCOMO
CA 1-3-20	1 + 3 + 20	Vodafone
CA 1-5-7	1 + 5 + 7	LGU +
CA 1-7-20	1 + 7 + 20	Vodafone
CA 1-19-21	1 + 19 + 21	NTT DOCOMO
CA 1-42-42	1 + 42 + 42	NTT DOCOMO
CA 2-2-13	2 + 2 + 13	Verizon Wireless
CA 2-4-4	2 + 4 + 4	TMO-US
CA 2-4-5	2 + 4 + 5	US Cellular
CA 2-4-13	2 + 4 + 13	Verizon Wireless
CA 2-5-12	2 + 5 + 12	US Cellular
CA 2-5-30	2 + 5 + 30	AT&T
CA 2-12-12	2 + 12 + 12	AT&T
CA 2-12-30	2 + 12 + 30	AT&T
CA 2-29-30	2 + 29 + 30	AT&T
CA 3-3-7	3 + 3 + 7	TeliaSonera
CA 3-7-7	3 + 7 + 7	Orange, Deutsche Telekom
CA 3-7-20	3 + 7 + 20	Vodafone
CA 4-4-12	4 + 4 + 12	TMO-US
CA 4-4-13	4 + 4 + 13	Verizon Wireless
CA 4-5-12	4 + 5 + 12	US Cellular
CA 4-5-30	4 + 5 + 30	AT&T
CA 4-12-12	4 + 12 + 12	AT&T
CA 4-12-30	4 + 12 + 30	AT&T
CA 4-29-30	4 + 29 + 30	AT&T
CA 19-42-42	19 + 42 + 42	NTT DOCOMO

Carrier Aggregation bands in 3GPP Rel-12

Inter-band 2 UL CA:

CA Band	E-UTRA operating band	Requested by
TBD	TBD in Dec 2014	

Intra-band non-contiguous 2 UL CA:

CA Band	E-UTRA operating band	Requested by
TBD	TBD in Dec 2014	

Figure 1c. 3GPP Carrier Aggregation Bands in Rel-12.

3GPP defined a maximum of five component carriers that can be aggregated. Hence, the maximum aggregated bandwidth is 100 MHz. CA can be used by both LTE frame structures; meaning by both FDD and TDD, and it can be enabled for both DL and UL direction.

Figure 2 illustrates the major differences between FDD and TDD modes. [5]

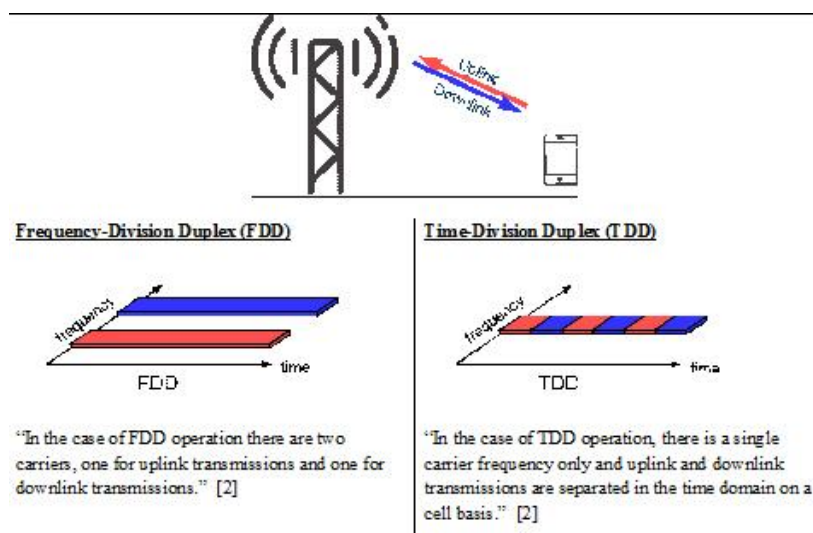


Figure 2. How FDD and TDD Differ.

In FDD, the number of aggregated carriers can be different in DL and UL, which is also referred to as asymmetric configuration. However, the number of UL component carriers is always equal to or lower than the number of DL component carriers. In the general case, the individual component carriers of DL and UL can also be of different bandwidths. However, in Rel-10, Rel-11 and Rel-12, there are limitations requiring the UL CC bandwidth to be the same as the DL CC bandwidth. The same requirements apply for TDD.

Figure 3 illustrates the different CA configurations [4].

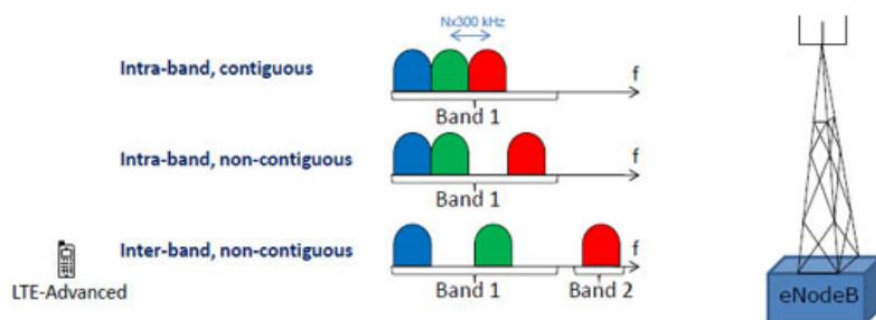


Figure 3. Intra-Band and Inter-Band Aggregation Alternatives.

The possible variety of CC aggregation combinations offers significant flexibility for efficient spectrum utilization and gradual re-farming of frequencies previously used by other systems, such as GSM and UMTS. The combination of component carriers is referred as CA configuration.

4.3 CONTIGUOUS AND NON-CONTIGUOUS INTRA-BAND CA

With the current maximum channel bandwidth in 3GPP limited to 20 MHz in LTE and LTE-Advanced, intra-band CA provides a path to wider bandwidths. Due to the commonality of the frequencies and propagation characteristics, intra-band CA is primarily used to enhance capacity rather than coverage. Intra-band CA can be contiguous, non-contiguous or a combination of contiguous and non-contiguous.

Contiguous CA Bandwidth Classes and Guard Bands

Contiguous CA combinations are defined in 3GPP TS 36.101 Table 5.6A-1 according to CA bandwidth classes. [2] The bandwidth class depends on the number of Physical Resource Blocks (PRBs) and the number of component carriers. The CA configuration indicates a combination of LTE operating band(s) and CA bandwidth class(es). The *CA bandwidth class* indicates a combination of maximum ATBC and maximum number of CCs, where ATBC is the *Aggregated Transmission Bandwidth Configuration* that is equal to the total number of aggregated PRBs. Guard bands are specifically defined for contiguous CA, based on the widest component carrier in CA configuration.

Table 2 presents the CA bandwidth classes and corresponding nominal guard bands.[2] As can be noted, a maximum of three CC's is already available, while a usage of four CCs is expected to be available in future releases after further study.

Table 2. CA Bandwidth Classes and Corresponding Nominal Guard Bands.

CA Bandwidth Class	Aggregated Transmission Bandwidth Configuration	Number of contiguous CC	Nominal Guard Band BW_{GB}
A	$N_{RB,agg} \leq 100$	1	$a_1 BW_{Channel(1)} - 0.5 \Delta f_1$ (NOTE 2)
B	$25 < N_{RB,agg} \leq 100$	2	$0.05 \max(BW_{Channel(1)}, BW_{Channel(2)}) - 0.5 \Delta f_1$
C	$100 < N_{RB,agg} \leq 200$	2	$0.05 \max(BW_{Channel(1)}, BW_{Channel(2)}) - 0.5 \Delta f_1$
D	$200 < N_{RB,agg} \leq 300$	3	$0.05 \max(BW_{Channel(1)}, BW_{Channel(2)}, BW_{Channel(3)}) - 0.5 \Delta f_1$
E	$[300] < N_{RB,agg} \leq [400]$	For Further Study (FFS)	FFS
F	$[400] < N_{RB,agg} \leq [500]$	FFS	FFS
NOTE 1: $BW_{Channel(j)}$, $j = 1, 2, 3$, is the channel bandwidth of an E-UTRA component carrier according to Table 5.6-1 and $\Delta f_1 = \Delta f$ for the downlink with Δf the subcarrier spacing while $\Delta f_1 = 0$ for the uplink.			
NOTE 2: $a_1 = 0.16/1.4$ for $BW_{Channel(1)} = 1.4$ MHz whereas $a_1 = 0.05$ for all other channel bandwidths.			

Channel Spacing in Intra-Band CA

Channel spacing between two or more adjacent contiguous aggregated carriers is defined by equation 1.

$$\text{Eq 1. Nominal channel spacing} = \left\lceil \frac{BW_{Channel(1)} + BW_{Channel(2)} - 0.1 |BW_{Channel(1)} - BW_{Channel(2)}|}{0.6} \right\rceil 0.3 \text{ [MHz]}$$

The 3GPP specifications also allow spacing that is narrower than nominal. However, in order to maintain orthogonality between 15 kHz OFDM tones and also maintain backward compatibility, the channel spacing must be an integer multiple of both 15 kHz and 100 kHz; i.e., 300kHz (Least Common Multiple $[15,100] = 300$). Minimal channel spacing for each carrier bandwidth has been defined. However, only nominal spacing has been verified and is guaranteed to work.

Complementary information to the CA configuration is the supported channel bandwidths per operating band and the applicable combinations of the aggregated CC according to the bandwidth usage for each CC. This information is given by the bandwidth combination set and is defined for each CA configuration. There could be more than one bandwidth combination set applicable per CA configuration. The actual supported bandwidth combination sets are indicated per supported band combination/CA configuration in the UE radio access capability.

To exemplify the naming convention of a CA configuration and the bandwidth combination set usage, let us relate to CA_1C configuration, which is available in 3GPP Rel-10. This CA configuration states that the UE can operate on Band 1, with two continuous components carriers, with a maximum of 200 RBs. The bandwidth combination set then states that the allocation of those 200 RBs can be either 75 RBs on both CCs or 100RBs on both CCs. Figure 4 shows the naming conventions.

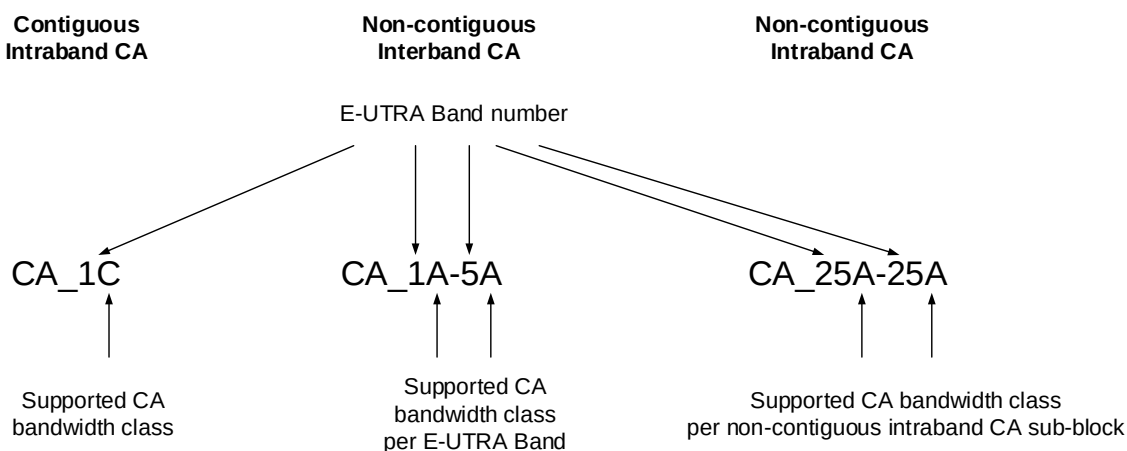


Figure 4. CA Configuration Naming Conventions

Table 3 presents the status of CA combinations for intra-band contiguous case as of June 2014. [2]

Table 3. LTE CA Configurations and Bandwidth Combination Sets Defined for Intra-Band Contiguous CA.

LTE CA configuration / Bandwidth combination set					
E-UTRA CA configuration	Component carriers in order of increasing carrier frequency			Maximum aggregated bandwidth [MHz]	Bandwidth combination set
	Allowed channel bandwidths for carrier [MHz]	Allowed channel bandwidths for carrier [MHz]	Allowed channel bandwidths for carrier [MHz]		
CA_1C	15	15		40	0
	20	20			
CA_3C	5, 10, 15	20		40	0
	20	5, 10, 15, 20			
CA_7C	15	15		40	0
	20	20			
	10	20		40	1
	15	15, 20			
	20	10, 15, 20			
CA_23B	10	10		20	0
	5	15			
CA_27B	1.4, 3, 5	5		13	0
	1.4, 3	10			
CA_38C	15	15		40	0
	20	20			
CA_39C	5,10,15	20		35	0
	20	5, 10, 15			
CA_40C	10	20		40	0
	15	15			
	20	10, 20			
CA_41C	10	20		40	0
	15	15, 20			
	20	10, 15, 20			
	5, 10	20		40	1
	15	15, 20			
	20	5, 10, 15, 20			
CA_40D	10, 20	20	20	60	0
	20	10	20		
	20	20	10		
CA_41D	10	20	15	60	0
	10	15, 20	20		

	15	20	10, 15		
	15	10, 15, 20	20		
	20	15, 20	10		
	20	10, 15, 20	15, 20		
CA_42C	5, 10, 15, 20	20	5, 10, 15, 20	40	0
	20	5, 10, 15	20		
NOTE 1: The CA configuration refers to an operating band and a CA bandwidth class specified in Table 5.6A-1 (the indexing letter). Absence of a CA bandwidth class for an operating band implies support of all classes.					
NOTE 2: For the supported CC bandwidth combinations, the CC downlink and uplink bandwidths are equal					

Table 4 presents the status of CA combinations for intra-band non-contiguous case as of June 2014. [2]

Table 4. LTE CA Configurations and Bandwidth Combination Sets Defined for Non-Contiguous Intra-Band CA.

LTE CA configuration / Bandwidth combination set				
LTECA configuration	Component carriers in order of increasing carrier frequency		Maximum aggregated bandwidth [MHz]	Bandwidth combination set
	Allowed channel bandwidths for carrier [MHz]	Allowed channel bandwidths for carrier [MHz]		
CA_2A-2A	5, 10, 15, 20	5, 10, 15, 20	40	0
CA_3A-3A	5, 10, 15, 20	5, 10, 15, 20	40	0
CA_4A-4A	5, 10, 15, 20	5, 10, 15, 20	40	0
CA_7A-7A	5	15	40	0
	10	10, 15		
	15	15, 20		
	20	20		
CA_23A-23A	5	10	15	0
CA_25A-25A	5, 10	5, 10	20	0
CA_41A-41A	10, 15, 20	10, 15, 20	40	0
	5, 10, 15, 20	5, 10, 15, 20	40	1
CA_41A-41C	5, 10, 15, 20	See Table 5.6A.1-1	60	0
CA_41C-41A	See Table 5.6A.1-1	5, 10, 15, 20	60	0
CA_42A-42A	5, 10, 15, 20	5, 10, 15, 20	40	0

4.4 INTER-BAND CA

Operators typically own spectrum in multiple bands. Thus inter-band CA is an efficient technology to combine their spectrum resources for increased data rates. In addition to providing operators with a powerful new tool to get the most out of their spectrum resources, inter-band CA also improves coverage in the scenario where the two component carriers are designed for varying coverage. Inter-band CA refers to aggregating component carriers belonging to different frequency bands. Each inter-band CA configuration is defined by a bandwidth combination set and maximum aggregated bandwidth.

For practical purposes, 3GPP limits the number of inter-band combinations defined to those required by operators. Table 5 gives the different inter-band combinations as defined in Rel-12 of the 3GPP specifications as of June 2014. [2]

Table 5. LTE CA Configurations and Bandwidth Combination Sets Defined for Inter-Band CA.

LTE CA configuration / Bandwidth combination set									
LTE CA Configuration	LTE Bands	1.4 MHz	3 MHz	5 MHz	10 MHz	15 MHz	20 MHz	Maximum aggregated bandwidth [MHz]	Bandwidth combination set
CA_1A-5A	1				Yes			20	0
	5				Yes				
	1			Yes	Yes	Yes	Yes	30	1
	5			Yes	Yes				
CA_1A-8A	1			Yes	Yes	Yes	Yes	30	0
	8			Yes	Yes				
	1			Yes	Yes			20	1
	8			Yes	Yes				
CA_1A-11A	1			Yes	Yes	Yes	Yes	30	0
	11			Yes	Yes				
CA_1A-18A	1			Yes	Yes	Yes	Yes	35	0
	18			Yes	Yes	Yes			
	1			Yes	Yes			20	1
	18			Yes	Yes				
CA_1A-19A	1			Yes	Yes	Yes	Yes	35	0
	19			Yes	Yes	Yes			
CA_1A-20A	1			Yes	Yes	Yes		25	0
	20			Yes	Yes				
CA_1A-21A	1			Yes	Yes	Yes	Yes	35	0
	21			Yes	Yes	Yes			
CA_1A-26A	1			Yes	Yes	Yes	Yes	35	0
	26			Yes	Yes	Yes			
	1			Yes	Yes			20	1
	26			Yes	Yes				
CA_2A-4A	2	Yes	Yes	Yes	Yes	Yes	Yes	40	0
	4			Yes	Yes	Yes	Yes		
	2			Yes	Yes			20	1
	4			Yes	Yes				

CA_2A-5A	2			Yes	Yes	Yes	Yes	30	0
	5			Yes	Yes				
CA_2A-12A	2			Yes	Yes	Yes	Yes	30	0
	12			Yes	Yes				
CA_2A-13A	2			Yes	Yes	Yes	Yes	30	0
	13				Yes				
	2			Yes	Yes			20	1
	13				Yes				
CA_2A-17A	2			Yes	Yes			20	0
	17			Yes	Yes				
CA_2A-29A	2			Yes	Yes			20	0
	29		Yes	Yes	Yes				
	2			Yes	Yes			20	1
	29			Yes	Yes				
	2			Yes	Yes	Yes	Yes	30	2
	29			Yes	Yes				
CA_3A-5A	3				Yes	Yes	Yes	30	0
	5			Yes	Yes				
	3				Yes			20	1
	5			Yes	Yes				
	3			Yes	Yes	Yes	Yes	30	2
	5			Yes	Yes				
CA_3A-7A	3			Yes	Yes	Yes	Yes	40	0
	7				Yes	Yes	Yes		
CA_3A-8A	3				Yes	Yes	Yes	30	0
	8			Yes	Yes				
	3				Yes			20	1
	8			Yes	Yes				
CA_3A-19A	3			Yes	Yes	Yes	Yes	35	0
	19			Yes	Yes	Yes			
CA_3A-20A	3			Yes	Yes	Yes	Yes	30	0
	20			Yes	Yes				
	3			Yes	Yes	Yes	Yes	40	1
	20			Yes	Yes	Yes	Yes		
CA_3A-26A	3			Yes	Yes	Yes	Yes	35	0
	26			Yes	Yes	Yes			
	3			Yes	Yes			20	1
	26			Yes	Yes				
CA_3A-27A	3			Yes	Yes	Yes	Yes	30	0
	27			Yes	Yes				
CA_3A-28A	3			Yes	Yes	Yes	Yes	40	0
	28			Yes	Yes	Yes	Yes		
CA_4A-5A	4			Yes	Yes			20	0
	5			Yes	Yes				
	4			Yes	Yes	Yes	Yes	30	1
	5			Yes	Yes				
CA_4A-7A	4			Yes	Yes			30	0

	7			Yes	Yes	Yes	Yes		
CA_4A-12A	4	Yes	Yes	Yes	Yes			20	0
	12			Yes	Yes				
	4	Yes	Yes	Yes	Yes	Yes	Yes	30	1
	12			Yes	Yes				
CA_4A-13A	4			Yes	Yes	Yes	Yes	30	0
	13				Yes				
	4			Yes	Yes			20	1
	13				Yes				
CA_4A-17A	4			Yes	Yes			20	0
	17			Yes	Yes				
CA_4A-29A	4			Yes	Yes			20	0
	29		Yes	Yes	Yes				
	4			Yes	Yes			20	1
	29			Yes	Yes				
	4			Yes	Yes	Yes	Yes	30	2
	29			Yes	Yes				
CA_5A-7A	5	Yes	Yes	Yes	Yes			30	0
	7				Yes	Yes	Yes		
CA_5A -12A	5			Yes	Yes			20	0
	12			Yes	Yes				
CA_5A-17A	5			Yes	Yes			20	0
	17			Yes	Yes				
CA_5A-25A	5			Yes	Yes			30	0
	25			Yes	Yes	Yes	Yes		
CA_7A-20A	7				Yes	Yes	Yes	30	0
	20			Yes	Yes				
	7				Yes	Yes	Yes	40	1
	20			Yes	Yes	Yes	Yes		
CA_7A-28A	7			Yes	Yes	Yes	Yes	35	0
	28			Yes	Yes	Yes			
CA_8A-20A	8			Yes	Yes			20	0
	20			Yes	Yes				
	8		Yes	Yes	Yes			20	1
	20			Yes	Yes				
CA_8A-40A	8			Yes	Yes			30	0
	40			Yes	Yes	Yes	Yes		
CA_11A-18A	11			Yes	Yes			25	0
	18			Yes	Yes	Yes			
CA_12A-25A	12			Yes	Yes			30	0
	25			Yes	Yes	Yes	Yes		
CA_19A-21A	19			Yes	Yes	Yes		30	0
	21			Yes	Yes	Yes			
CA_20A-32A	20			Yes	Yes			30	0
	32			Yes	Yes	Yes	Yes		
CA_23A-29A	23			Yes	Yes	Yes	Yes	30	0
	29		Yes	Yes	Yes				

	23			Yes	Yes			20	1
	29		Yes	Yes	Yes				
CA_39A-41A	39				Yes	Yes	Yes	40	0
	41						Yes		

NOTE 1: The CA Configuration refers to a combination of an operating band and a CA bandwidth class specified in Table 5.6A-1 (the indexing letter). Absence of a CA bandwidth class for an operating band implies support of all classes.

NOTE 2: For each band combination, all combinations of indicated bandwidths belong to the set.

NOTE 3: For the supported CC bandwidth combinations, the CC downlink and uplink bandwidths are equal.

5. CA TECHNOLOGY DESCRIPTION

5.1 CA BENEFITS AND PERFORMANCE

CA is an effective tool to combine together fragmented operator spectrum and deliver higher throughputs with a bigger spectrum pipe. However, CA has several additional benefits:

- **Higher speeds:** Aggregation of carriers increases spectrum resources, which provides higher speeds across the cell coverage.
- **Capacity gain:** Aggregating multiple carriers increases spectrum but also includes trunking gains from dynamically scheduling traffic across the entire spectrum. This in turn increases cell capacity and network efficiency and improves the experience for all users. A user previously facing congestion on one carrier can be seamlessly scheduled on a carrier with more capacity and maintain a consistent experience.
- **Optimum utilization of an operator's spectrum resources:** The majority of operators have fragmented spectrum covering different bands and bandwidths, Carrier Aggregation helps combine these into more valuable spectrum resource.
- **Aggregation of spectrum resources in HetNet or LTE-Unlicensed scenarios generates new spectrum sources for operators.** Licensed spectrum is in perpetually short supply, and even when it is available, it is sometimes an expensive investment. CA enables operators to maximize the return on those investments, as well as use unlicensed spectrum to supplement their portfolios.

The CA feature increases the bandwidth for a CA-capable UE by aggregating several LTE carriers, thereby increasing the UE's bit rate. Each of the aggregated carriers is referred to as a CC. In order to keep backward compatibility, the aggregation is based on Rel-8/Rel-9 carrier structure and so legacy UEs should still be able to coexist with LTE-Advanced on at least part of the total bandwidth. Therefore, the component carrier can have a bandwidth of 1.4, 3, 5, 10, 15 or 20 MHz.

Even though LTE Rel-8 can support bandwidths up to 20 MHz, most American wireless operators do not have that much contiguous spectrum. In spectrum below 2 GHz, most operators have 5-15 MHz of contiguous spectrum in a single frequency band. Also, many operators own the rights to use spectrum in many different bands. So from a practical perspective, CA offers operators a path to combine spectrum assets within the bands they operate in and to combine assets across multiple frequency bands.

Figure 5 shows the enhanced user experience that can be provided with CA by combining a single AWS carrier with a single 700 MHz carrier. Under lightly loaded conditions, CA devices can better utilize resources on both component carriers, rather than being restricted to a single carrier block of spectrum.

In this example, two carriers of spectrum are combined together to nearly double the median end user throughput.

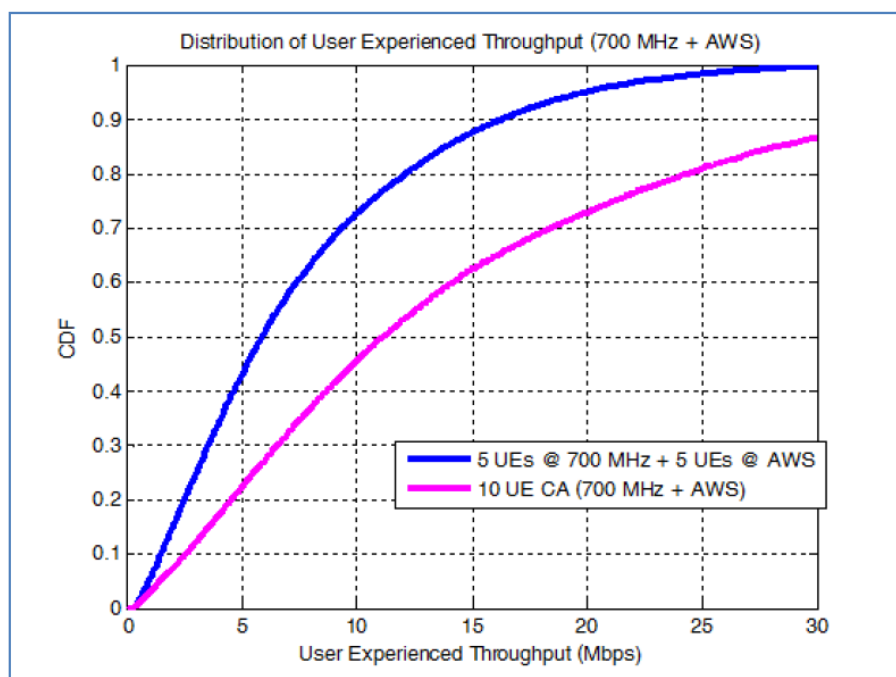


Figure 5. User Performance Benefits of CA [6].

Figure 6 further presents some lab results from CA testing showing the improvements of DL throughput due to CA.

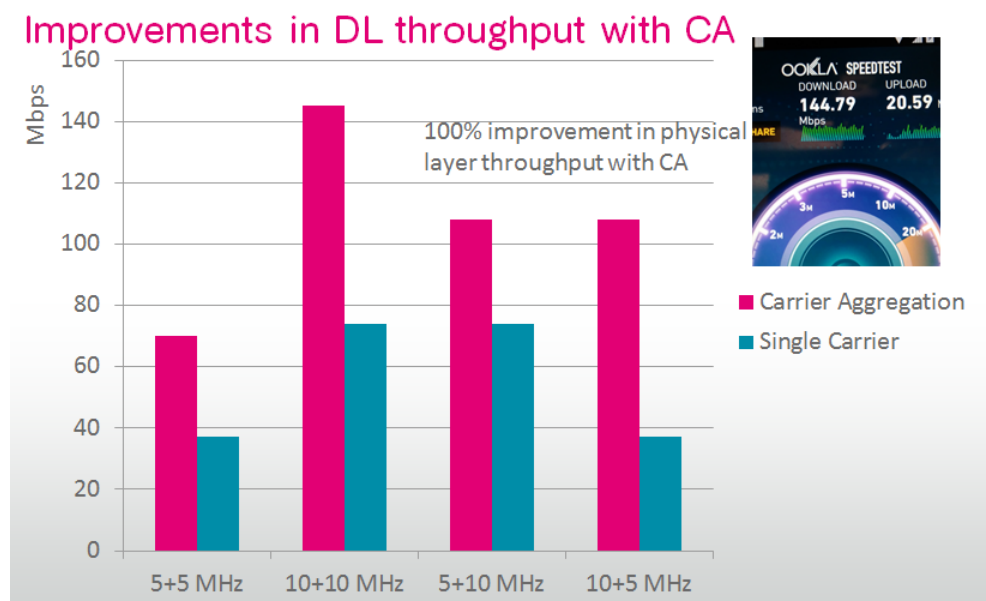


Figure 6. Example of CA Lab Results from a U.S. Operator.

5.2 CELL MANAGEMENT AND CELL ACTIVATION-DEACTIVATION

To efficiently utilize the multiple component carriers for data transmission, additional procedures, including cell management and cell activation/deactivation, are designed for CA.

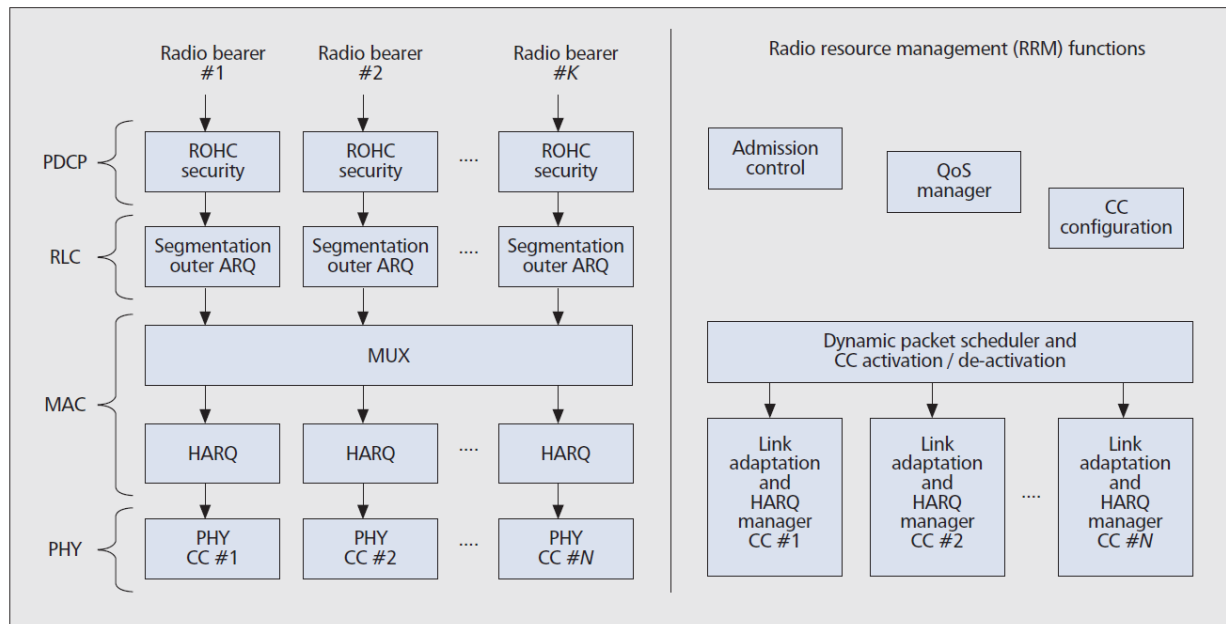


Figure 7. Overview of the DL User Plane Architecture (left side) and the Corresponding Radio Resource Management (RRM) Algorithms (right side) [1].

In LTE-Advanced CA, a CC used for data transmission is also referred to as a serving cell and is also treated as such by the higher layer procedures.

For FDD, a serving cell comprises a pair of different carrier frequencies for DL and UL transmissions. For TDD, a serving cell is defined for a single carrier frequency where DL and UL transmissions occur in different transmission time intervals. In LTE-A, where number of DL and UL CCs could be asymmetric, the relation between DL and UL CCs may be signalled in the system information block (SIB) type 2 and is common for all UEs.

Also, for each UE there is a single serving cell that provides all necessary control information and functions, such as Non-Access Stratum (NAS) mobility information, Radio Resource Control (RRC) and connection maintenance. This serving cell is referred to as the primary cell (PCell). In the DL, the carrier corresponding to the PCell is the Downlink Primary Component Carrier (DL PCC), while in the UL it is the Uplink Primary Component Carrier (UL PCC).

Other serving cells are referred to as secondary cells (SCells) and are used for bandwidth expansion for the particular UE. It is to be noted that there are additional differences in the functionality.

The cell where an initial access is performed by the UE is the cell which is related by the network as the PCell. Changing of a PCell can be performed only via a handover procedure. The network can configure additional CCs as SCells only for a CA-capable device with an RRC connection on a PCell. The configuration of SCells is done via dedicated RRC signalling to the UE, as well as any addition, reconfiguration or removal of SCells.

CA offers a cell activation/deactivation mechanism for configured SCells to optimize and reduce UE power consumption. Activating/deactivating of a SCell refers to activating/deactivating the CC of the SCell, while activation of an SCell means applying normal SCell operation, which are detailed in [7]. For a deactivated SCell, as detailed in TS36.321 Rel-11, the UE does not transmit SRS² on the SCell, does not report CQI/PMI/RI/PTI³ for the SCell, does not transmit on UL-SCH⁴ or on RACH⁵ on the SCell and does not monitor the PUCCH on the SCell nor for the SCell.

The activation/deactivation of the SCells is performed by the network independently for each of the SCells serving the UE, according to internal algorithms of the network (aiming, for example to meet the UE's current traffic demand). The configured SCells are initially deactivated upon addition and after a handover.

Activation is done via a dedicated Medium Access Control (MAC) control element signalling, while deactivation can be done either via the same dedicated MAC control element (sets differently) signalling or according to a dedicated timer maintained by the UE and the network for each SCell. The Pcell is active for a UE in RRC connected mode and it cannot be deactivated.

5.3 CA SCHEDULING

The 3GPP has defined two main alternatives for CA scheduling: either resources are scheduled on the same carrier on which the scheduling grant (over PDCCH)⁶ is received, or resources are scheduled on a different carrier than the carrier on which scheduling grant is received.

For the first alternative, the PDCCH is separately coded for each CC that serves the UE (PCell and SCells) and reuses Rel-8/9 PDCCH structure including Rel-8/9 DCI⁷ formats.

For the second alternative, a single PDCCH available at the Pcell assigns resources for all CC's of the UE. The second option is referred as a cross-carrier scheduling, and its main use case is for heterogeneous networks (further information is available in Section 6.4). To implement this method, a 3

² SRS refers to Sounding Reference Signals, which are transmitted on the UL containing a known sequence to both network and devices and allow the network to estimate the quality of the channel at different frequencies. SRS may be transmitted periodically in a wider bandwidth (beyond the UL resources blocks allocated for UL data transmission) and also when there is no UL data transmission, so the channel information obtained from SRS is a good input to the UL scheduler at the eNB.

³ CQI is Channel Quality Indicator. PMI stands for Pre-coding Matrix Indicator and RI is Rank Indicator. These three values are computed on the fly in LTE systems and used to try to optimize resource allocation among the various UEs that are requesting service. The CQI is reported by UE to eNB. UE indicates modulation scheme and coding scheme to eNB, that would allow it to demodulate and decode the transmitted DL data with a maximum block error rate of 10 percent. The CQI reporting can be based on PMI and RI.

For PMI, the UE indicates to eNB which precoding matrix from a predefined codebook should be used for DL transmission by the eNB, which is determined by RI.

For RI the UE indicates to eNB, the number of spatial layers that should be used for downlink transmission to the UE.

PTI stands for precoding type indicator and it is a new feedback indicator added for LTE-Advanced, which is reported by the UE and allows distinguish slow from fast fading environments.

⁴ UL-SCH is a MAC uplink transport channel and is carried over the Physical Uplink Shared Channel. The UL-SCH is used for UL transmission of UE according to scheduling guidelines given by the eNB.

⁵ RACH – Random Access Channel. This channel is used for the execution of the Random Access Procedure in which the network (eNB) for the first time knows that some UE is trying to get access.

⁶ PDCCH stands for physical downlink control channel, which carries the layer one control.

⁷ DCI stands for Downlink scheduling Control Information and is transmitted through the Physical Downlink Control Channel (PDCCH) to support the transmission of the DL and UL transport channels for UEs associated with the eNB. DCI contains the resource allocation (MCS, assignment of resource block), TPC commands, HARQ information and MIMO precoding information.

bit Carrier Indicator Field, referred as CIF, indicates which CC the PDSCH⁸ allocation in the PDCCH is for.

Note that assignment of DL/UL data resources is never done using the cross-carrier scheduling method.

Figure 8 is taken from 3GPP and illustrates the two scheduling methods that can be in used when CA is applied. [4]

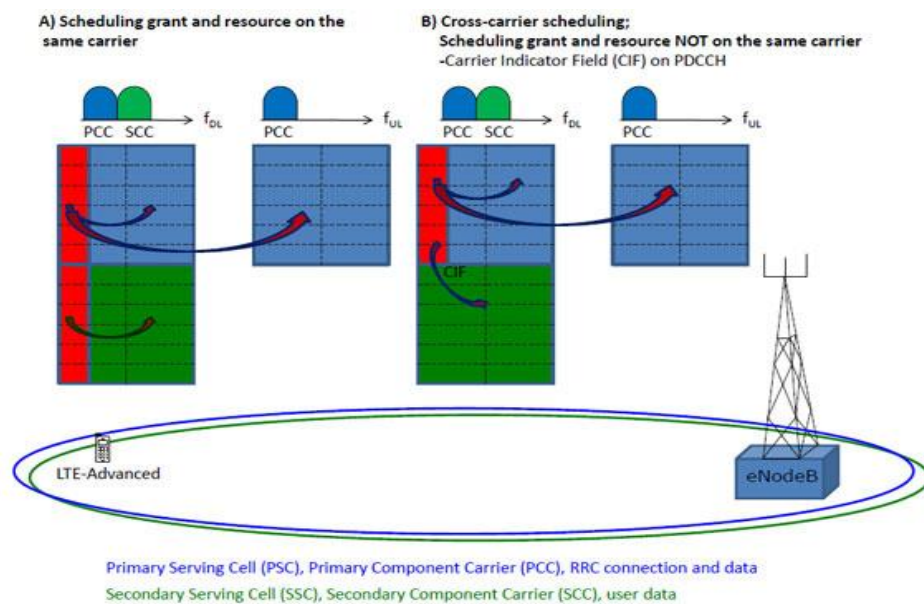


Figure 8. Scheduling Alternatives for CA (FDD Configuration).

5.4 MOBILITY SCENARIOS WITH CA

In LTE networks today, mobility is based on event-triggered measurements that enable handovers to a neighbor cell. The device reports measurement information in accordance with the measurement configuration as provided by E-UTRAN. As shown in Table 6, 3GPP Release 8 defines the following measurement events that can be configured to trigger mobility.

Table 6. Event Triggers for Handovers.

Event A1	Serving becomes better than absolute threshold
Event A2	Serving becomes worse than absolute threshold
Event A3	Neighbor becomes offset better than serving
Event A4	Neighbor becomes better than threshold
Event A5	Serving becomes worse than threshold1 and neighbor becomes better than threshold2
Event B1	Inter RAT neighbor becomes better than threshold
Event B2	Serving becomes worse than threshold1 and inter RAT neighbor becomes better than threshold2

⁸ The PDSCH is the channel that carries all user data and all signaling messages.

In a CA scenario, mobility is primarily based on PCell coverage and existing measurement events as defined above. A change of PCell always requires handover, while change of SCell can be performed through a reconfiguration procedure. During a handover procedure, CA or SCell is de-configured prior to handover, and a new SCell may be configured for the UE in the target eNB. De-configuration of the old SCell and configuration of a new SCell is through the RRC Reconfiguration procedure.

For example, consider the inter-band CA (Band 1: B1, Band 2: B2) and mobility scenario illustrated in Figure 9, where:

1. A CA-capable device has the PCC as B1 and SCC as B2.
2. As the UE moves towards poor coverage on B1, inter-frequency measurements are started on B2. Inter-frequency handover is triggered with B2 as PCC. B1 may be configured as SCC based on scheduler design.
3. Intra-frequency handover to neighbor cell B2. B2 continues as PCC.
4. B1 may be added as SCC, and CA can be scheduled.

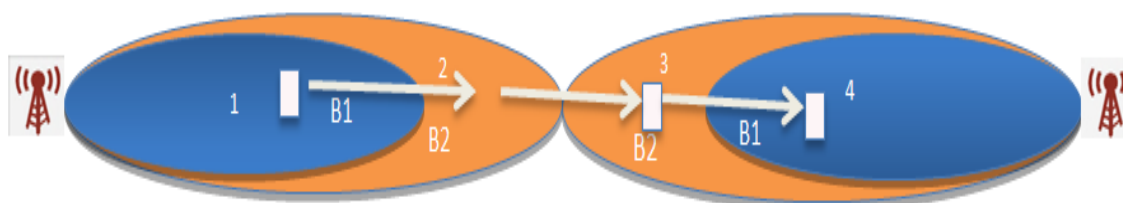


Figure 9. Inter-Band CA Handover Example.

3GPP Release 10 also introduces a new measurement event: Event A6 for detecting and reporting neighbor cells on the same frequency as SCell that becomes better than SCell. This enables selecting the best SCell for a given PCell and maximizes CA's benefits. Event A6 measurements are started at SCell configuration and can trigger the configuration of a new SCell and de-configuration of existing SCell.

5.5 UL AND DL AGGREGATION

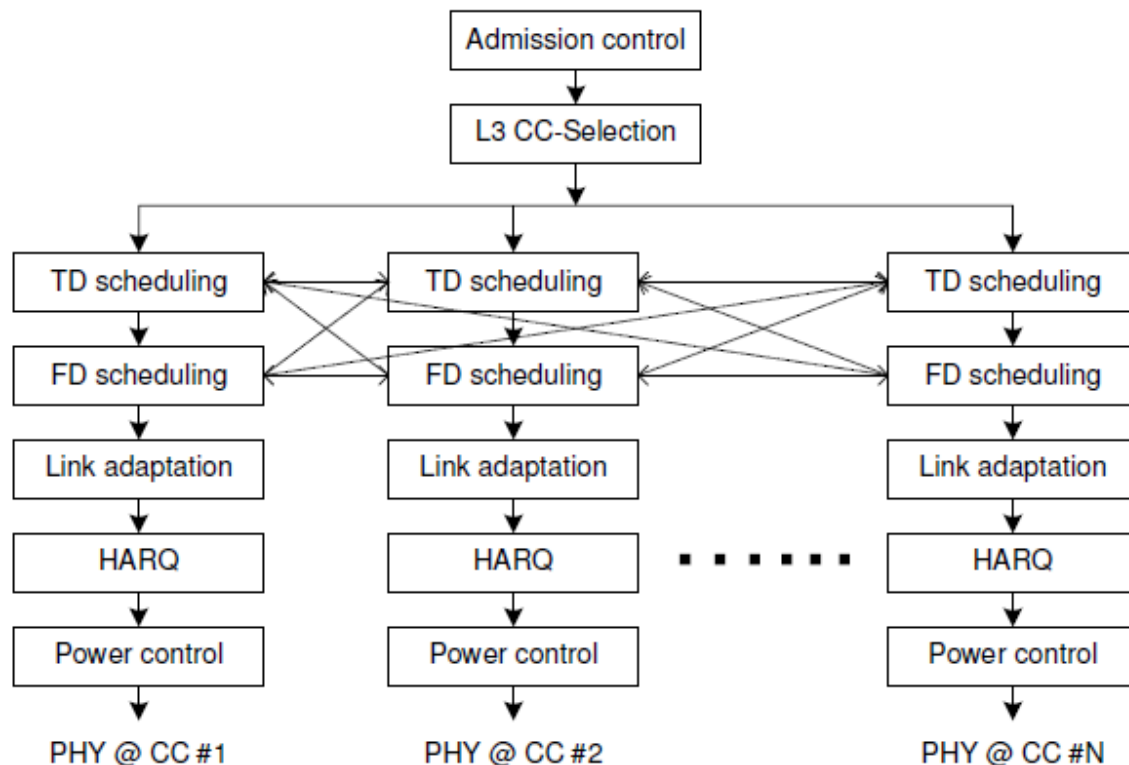
3GPP signaling supports up to 5 component carriers in any CA scheme. However, current CA proposals aggregate only up to three carriers. CA_40D and CA_41D already deliver up to 60 MHz DL bandwidth to the UE for three-carrier aggregation. In 3GPP, the UL and DL CA bands are signaled separately, making UL CA independent of DL CA. An example is CA_41D, which aggregates three DL carriers contiguously in Band 41 (2.5GHz band) and is being developed for DL only. CA_41D can be deployed along with single-carrier B41 on the UL, or it can be deployed with two-carrier CA_41C on the UL.

5.6 UL CA [8]

Though there are a few similarities between UL and DL in CA, some substantial differences exist. One of the main differences in UL compared to DL is a UE's transmission power limit. In DL, allocating more CCs to a UE generally results in a higher throughput thanks to the larger bandwidth and higher transmission power.

However, this is not always the case in UL. Increasing the bandwidth does not necessarily result in an increase of data rates if a UE reaches its maximum transmission power. Moreover, the effects of

increased peak-to-average power ratio (PAPR) and inter-modulation can further result in a reduction of the UE maximum transmission power when a UE is simultaneously transmitting over multiple CCs. Such cost might counterbalance the gain brought by multiple-CC transmission and even result in a performance loss. So it is not a good idea to always allocate multiple CCs to LTE-Advanced UEs, especially to those power-limited LTE-Advanced UEs that experience unfavorable channel conditions.



The following figures show the uplink performance when considering an LTE-Advanced system consisting of two contiguous component carriers, each of which has 20 MHz bandwidth to form a wide band of 40 MHz. Figure 11 shows the 5-percentile cell edge user throughput versus the offered load in different scenarios. It is shown that the coverage performance of LTE-Advanced UEs is the same as that of Rel-8 UEs regardless of the setting of power back-off. In other words, there is no gain in coverage by using CA in uplink. This is because at the cell edge, UEs usually experience high path loss and are limited by the maximum transmission power. Even if those cell edge LTE-Advanced UEs are assigned to multiple CCs, and they do not have sufficient power to explore the increased transmission bandwidth. Therefore, cell

edge LTE-Advanced UEs are assigned to only one CC, which results in the same coverage and performance compared with Rel-8 UEs.

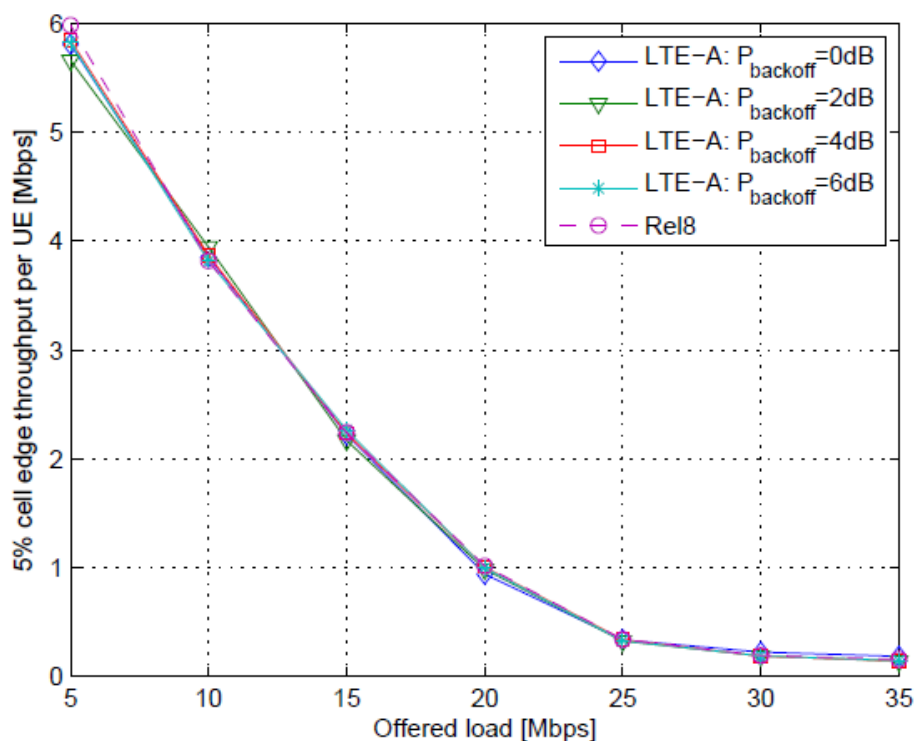


Figure 11. 5-Percentile Cell Edge User Throughput under Different Traffic Loads in Different Scenarios.

Figure 12 shows the average user throughput versus the offered load in different scenarios. The average user throughput for LTE-Advanced UEs can always achieve better performance than Rel-8 UEs regardless of the setting of power back-off. In one possible CC selection algorithm, power-limited LTE-Advanced UEs are assigned to one CC and therefore will not experience any loss from being scheduled over multiple CCs. Meanwhile LTE-Advanced UEs not operating close to their maximum transmission power are assigned to both CCs so that they can benefit from CA's advantages. The gain is high in low load but decreases as the offered load increases.

For example, the average throughput gain with CA is 64 percent in low load (62 percent PRB utilization) and 41 percent in high load (98 percent PRB utilization) for the case of $P_{\text{backoff}} = 4$ dB. The PRB utilization versus the offered load is depicted in Figure 13. For a given offered load, the average throughput of LTE-Advanced UEs decreases with the increase of power back-off. This is because the UL is power limited. The increase of power back-off will cause further reduction of the maximum UE transmission power, which results in a decreasing probability of being assigned to multiple CCs for LTE-Advanced UEs.

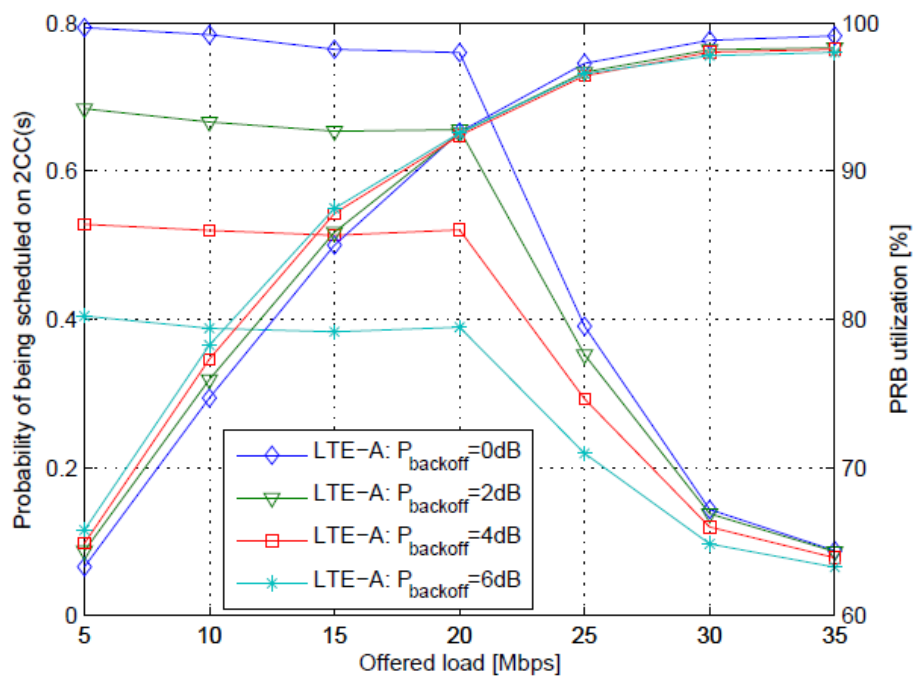


Figure 12. Probability of Being Scheduled on Both CCs Simultaneously and the Average PRB Utilization under Different Traffic Loads in Different Scenarios.

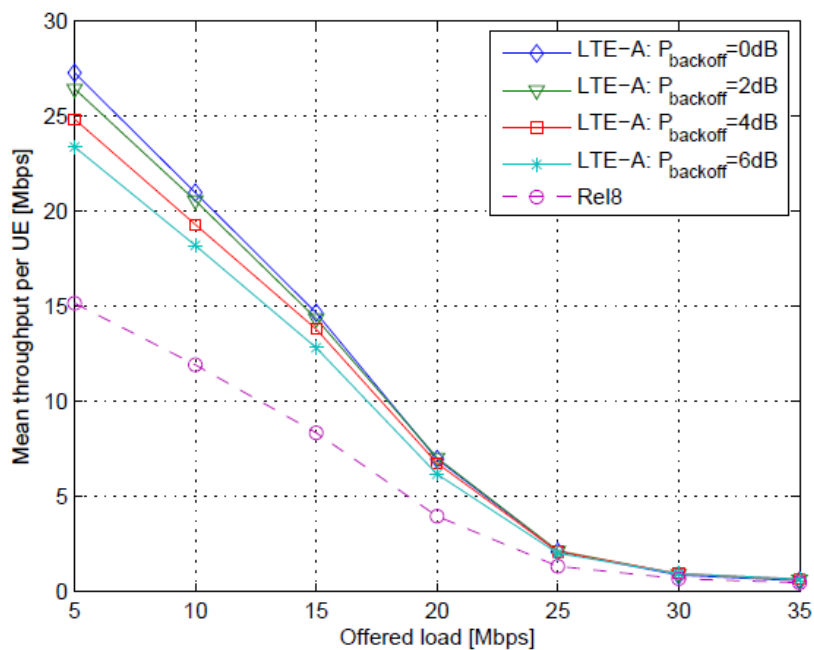


Figure 13. Average User Throughput under Different Traffic Loads in Different Scenarios.

6. CA DEPLOYMENT SCHEMES

6.1 INTRA-BAND CA

Both contiguous and non-contiguous intra-band CA cases are being deployed as discussed in the following examples.

6.1.1. EXAMPLE OF CONTIGUOUS INTRA-BAND CA USE CASE: CA_41D

CA_41D aggregates three DL carriers for enhanced capacity, offering up to 60 MHz of bandwidth. The component carriers must use the nominal spacing defined in Eq1 in Section 4.3 in order to provide internal guard bands. Since all component carriers are processed in the same UE receiver, the received power difference between carriers will be limited. The actual DL transmission bandwidth of each 20 MHz carrier is 18.015 MHz. Figure 14 illustrates this arrangement.

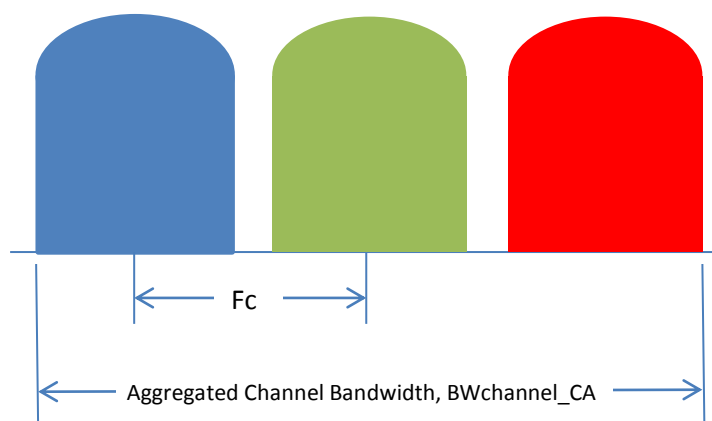


Figure 14. Three Aggregated DL Carriers.

NOTE: BWchannel_CA includes guard bands

6.1.2 EXAMPLE OF NON-CONTIGUOUS INTRA-BAND CA USE CASE: CA_41C-41A

CA_41C-41A aggregates three DL carriers for enhanced capacity by combining the two-carrier contiguous CA_41C with a single non-contiguous third carrier. In this specific use case, the sub-block gap can be very small because there are deployments that do not have an interferer in the sub-block gap. There are no specific guard band requirements for non-contiguous CA. However, contiguous CA sub-blocks that are part of the overall non-contiguous combination must meet the appropriate requirements. Figure 15 shows an example of CA_41C-41A with three 20 MHz carriers and a 4.8 MHz sub-block gap.

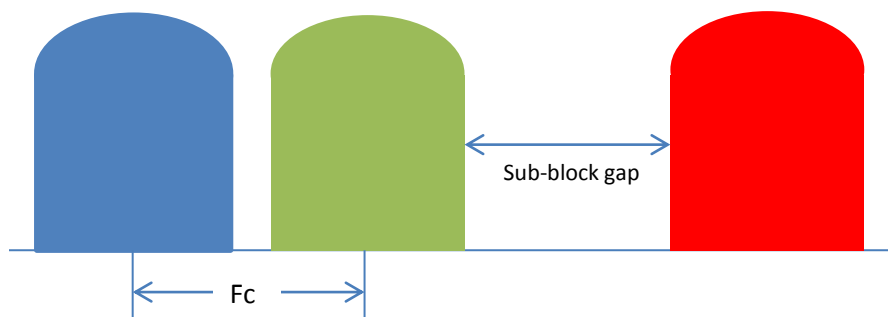


Figure 15. Three Aggregated 20 MHz Carriers with a 4.8 MHz Sub-Block Gap.

CA schemes may also combine contiguous and non-contiguous elements. One example is the CA_41C_41A scheme, which combines contiguous CA (CA_41C) with a non-contiguous element (41A) for a three-carrier DL CA capable of delivering up to 60 MHz DL bandwidth to the UE. Any CA combination that includes a non-contiguous element is considered non-contiguous CA. However, the contiguous element (i.e., CA_41C) must meet the requirements of contiguous CA for guard bands, spacing, etc. A non-contiguous element is characterized by a sub-block gap. For non-contiguous CA with potential interferers in the sub-block gap, a minimal sub-block gap is defined. For non-contiguous CA without potential interferers, a minimum gap is not defined.

6.2 INTER-BAND CA

Some examples of Inter-band CA cases are given in the following sections.

6.2.1. LOW AND HIGH BAND CA

This is a common configuration for carriers as many operators have valuable low band spectrum that has definite coverage advantages. Pairing this with other spectrum maximizes an operator's spectrum assets for coverage and capacity.

In North America, a low band such as 700 MHz paired with a higher band such as AWS is a common configuration and use case for CA to provide capacity and coverage benefits. A common concern in this scenario is the coverage and site density mismatch between the low and high bands limits the operation of CA. In case of a coverage mismatch, CA is not assigned in the areas where coverage does not overlap, as in Figure 16.

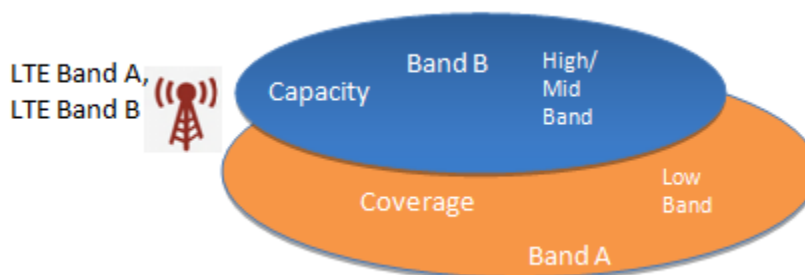


Figure 16. Coverage Mismatch's Effect on CA.

Site deployment for the low band may not overlap with site deployment for the high band and leaves pockets of the network with no CA. For example, Figure 17 illustrates the ratio of site deployment of the low band:high band as 1:2.

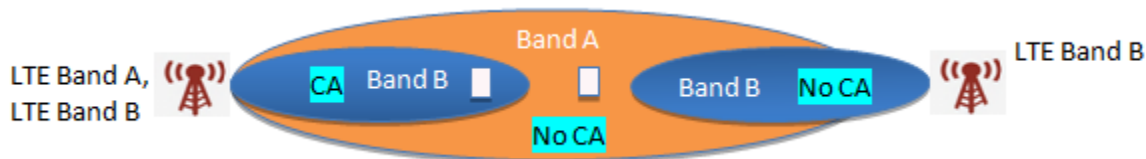


Figure 17. High and Low Band Sites Affect CA Availability.

Low and high band CA also has harmonics concerns where in CA mode, transmitting on the low band (for example, 700 MHz in Figure 18) generates a third harmonic (2100 MHz) that falls on the Band B receive path and impacts DL performance.

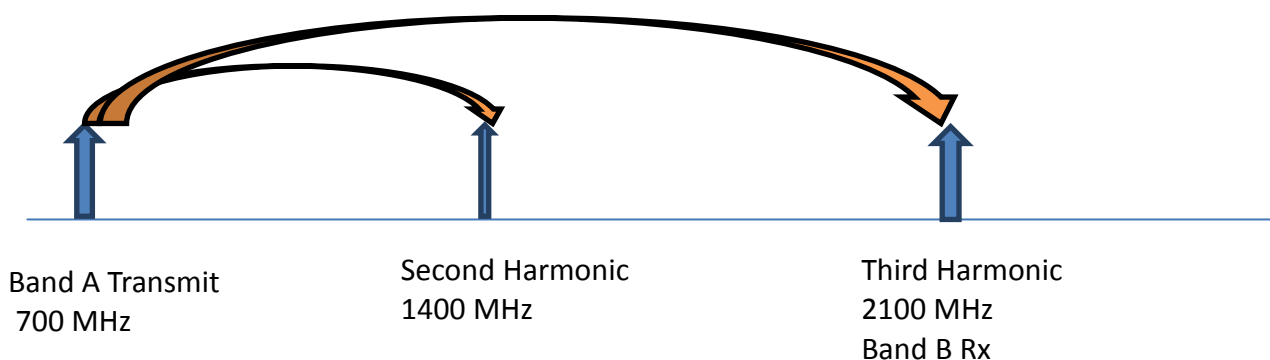


Figure 18. Third Harmonic Affects CA Availability.

3GPP has developed allowance in the standards for desensitization effects of harmonics in such scenarios. Other possible solutions to reduce harmonics concerns during CA include:

- Avoiding use of the primary carrier on low band.
- Advancements in RF and filter technology to mitigate harmonic RF effects.

CA_4A-12A is an example of a low and high band CA case that aggregates two DL CCs from Band 4 and Band 12 to provide up to 30 MHz of aggregated bandwidth. In this specific use case, Band 12 (700 MHz) provides coverage advantages over Band 4, and this combination of CA can be used to couple the coverage benefits of Band 12 with capacity and higher throughputs from additional spectrum of Band 4.

6.2.2. LOW AND LOWER BAND CA

Some operators have spectrum holdings that include more than one lower frequency band, such as 700 MHz band and 850 MHz. It is logical to conclude that these operators would also want to use inter-band CA techniques to provide their subscribers with an enhanced bandwidth experience using their various fragmented low band spectrum allocations. Although this scenario would not have (or at least not to the same degree) the problem of different cell sizes for the two (or more) aggregated channels, as experienced in the low band and high band scenario described above, other technical challenges arise.

In an effort to be efficient and to reduce handset size and weight, manufacturers of handsets (and chipsets) reuse various internal hardware components across many similar mobile bands. Antennas, diplexers, linear amplifiers and other RF devices are shared between bands. In order to support the CA of two low band frequencies, it may be necessary to build devices that require additional RF hardware to separate the two radio signal paths. This additional hardware may result in devices with increased weight, size, complexity and cost.

The low band and high band case discussed in the previous section does not have this limitation as the handheld device's RF components are probably already separate for the two bands.

In addition, the handset and chipset vendors, as well as the manufacturers of base station equipment, must consider tight filtering and other interference mitigation techniques in their designs. These features reduce cross-modulation and intermodulation, which could occur with multiple transmitters/receivers being operated simultaneously in close proximity. These are examples of why operators should consult with their RAN infrastructure vendors, as well as their device and chipset manufacturers, when developing their CA implementation plans.

6.2.3. HIGH AND HIGHER BAND CA

In a similar case to the low and lower band example described above, some operators will want to use CA techniques to provide their subscribers with an enhanced bandwidth experience using their multiple high band spectrum holdings. For instance, an operator may want to aggregate its AWS and 2500 MHz spectrum.

The technical challenges here are similar to the low and lower band scenario. Reuse of RF components in the manufacture of the handheld device must be considered. Also, filtering in the handset and in the base station must be well designed to minimize interference between carriers and components. So once again, operators should consult with their RF infrastructure and their device vendors when developing their CA strategy.

CA_2A-4A is an example of a high and higher band CA case that aggregates two DL component carriers from Band 2 (PCS 1.9GHz) and Band 4 (AWS 1.7/2.1GHz) to provide 40 MHz (300 Mbps) of aggregated bandwidth. The combination also defines bandwidth combination set 1 that allows an aggregated bandwidth of 20 MHz (150 Mbps). As both bands are consistent in coverage, this combination is primarily for higher capacity and throughput vs coverage. In this specific use case aggregation of two high bands requires a quadplexer in the device as defined in Section 7.3.

6.3 FDD-TDD CA [5]

Previously, CA has been possible only between FDD and FDD spectrum or between TDD and TDD spectrum. 3GPP has finalized the work on TDD-FDD CA, which offers the possibility to aggregate FDD and TDD carriers jointly. The main target with introducing the support for TDD-FDD CA is to allow the network to boost the user throughput by aggregating both TDD and FDD toward the same UE. This will allow the network to boost the UE throughput independently from where the UE is in the cell (at least for DL CA).

TDD and FDD CA would also allow dividing the load more quickly between the TDD and FDD frequencies. In short, TDD-FDD CA extends CA to be applicable also in cases where an operator has spectrum allocation in both TDD and FDD bands. The typical benefits of CA – more flexible and efficient

utilization of spectrum resources – are also made available for a combination of TDD and FDD spectrum resources. The Rel-12 TDD-FDD CA design supports either a TDD or FDD cell as the primary cell.

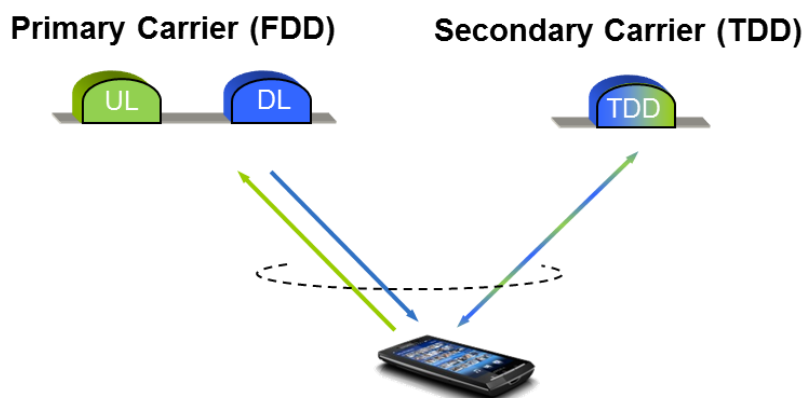


Figure 19. Example of TDD-FDD CA.

There are several different target scenarios in 3GPP for TDD-FDD CA, but there are two main scenarios that 3GPP aims to support. The first scenario assumes that the TDD-FDD CA is done from the same physical site that is typically a macro eNB. In the second scenario, the macro eNB provides either a TDD and FDD frequency, and the other frequency is provided from a Remote Radio Head (RRH) deployed at another physical location. The typical use case for the second scenario is that the macro eNB provides the FDD frequency and the TDD frequency from the RRH.

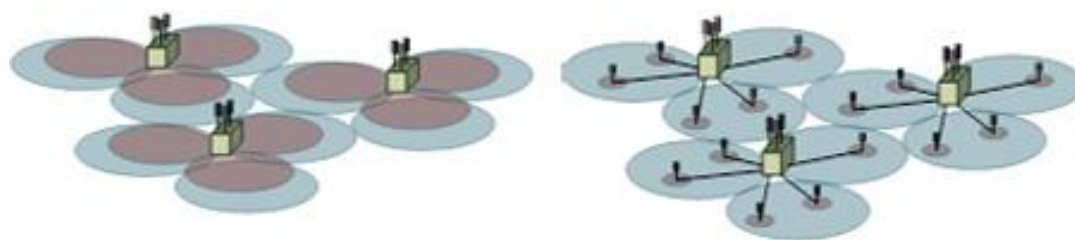


Figure 20. Collocated (left) and Non-Collocated (right) CA Deployments.

6.4 SUPPLEMENTAL DL [6]

Mobile broadband traffic will continue to grow exponentially over the coming years. The traffic is to a large extent asymmetrical with users typically downloading and streaming more data than they are uploading when engaged in day to day activities. However, in certain situations, such as sports events, public safety scenario or mass gatherings, the uplink traffic can become heavier than downlink, as spectators upload pictures, videos and files in social media and other services.

Because wireless broadband traffic tends to be asymmetrical, it is anticipated that mobile operators could use TD-LTE with different Uplink/Downlink subframe configurations or supplemental DL spectrum to supplement their spectrum holdings in other bands. The technology allows the bonding of the usual DL with a supplemental DL channel(s) in the same or in a different band, into a single wider DL as shown in Figure 21 to enhance the DL capability of mobile broadband networks by enabling significantly faster downloads and supporting a much greater number of users with mobile or portable wireless devices. 700 MHz FDD Band 29 is an example of Supplemental DL band.

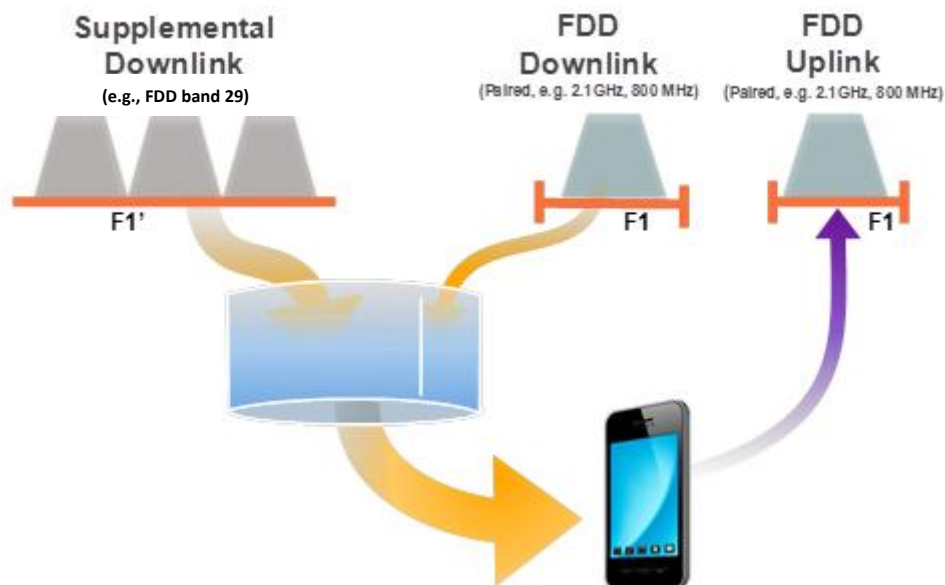


Figure 21. Mobile Supplemental DL.

6.5 HET NET SCENARIOS [9]

LTE-A Release 11 introduced the support of multi-site CA by supporting multiple UL Timing Advance enhanced UL power control for Heterogeneous Networks (HetNet) (Figure 22). In this case, the macro base station with full coverage could serve as Primary Cell (PCell) providing system information, RRC control signaling and bandwidth limited data transmission, while the small base station would serve local high data rate requirements. This CA technique will also be beneficial when macro and picocells are using dedicated carriers. Fast selection of small cells would be beneficial even for moving users as long as the proximity of the user within the small cell coverage can be detected.

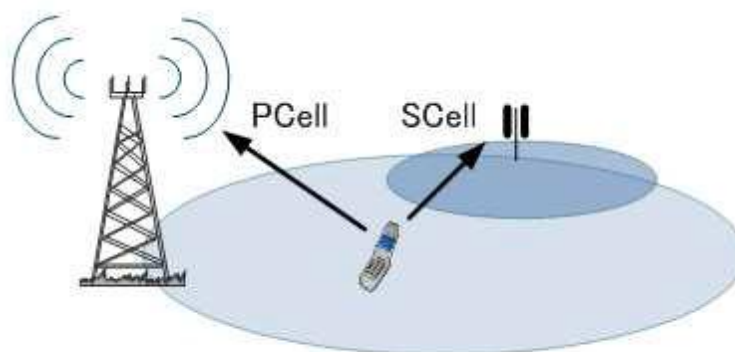


Figure 22. LTE-A HetNet Multi-site CA.

Nevertheless, there are drawbacks to this functionality. First, tight coordination is assumed between macro/pico layers, requiring a centralized architecture. Furthermore, in the case of UL CA, CA-capable terminals supporting Multiple Timing Advance are required. This will basically require two transmitters in the terminal, which is a big step in implementation complexity. At least initially, terminals having such

functionality will be in a minority. Later on, it might go hand in hand with the introduction of uplink LTE-A spatial multiplexing. However, this does not limit or impact the use of downlink CA in HetNet scenarios.

An alternative deployment is not necessarily based on a centralized architecture. The PDCCH protection is achieved with the other cell layers not transmitting PDCCH control signaling on the CC in question, for (at least) one CC in each cell layer. On the other hand, all CC are available for data allocation as signaled by the CC with the protected PDCHH.

Figures 23 illustrates the usage of cross carrier scheduling in a heterogeneous deployment. The high power cell layer transmits control signaling, PDCCH, on component carrier f_1 but not on component carrier f_2 . Meanwhile, for the low power cell layer, it is the opposite: The low power base station transmits PDCCH on component carrier f_2 but not on component carrier f_1 . The data itself occupies both CCs for both cell layer types (with or without Inter-Cell Interference Cancellation scheme enabled).

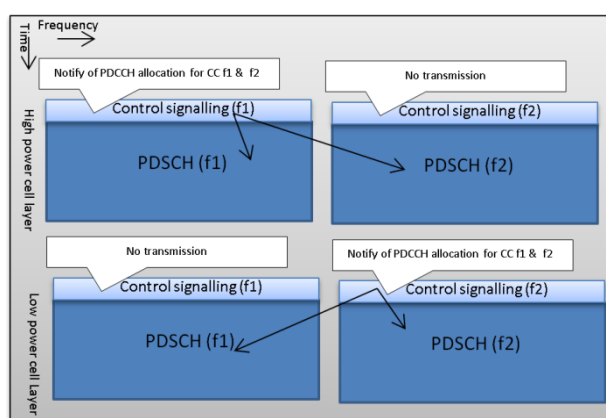


Figure 23. CC Scheduling in a Heterogeneous Deployment.

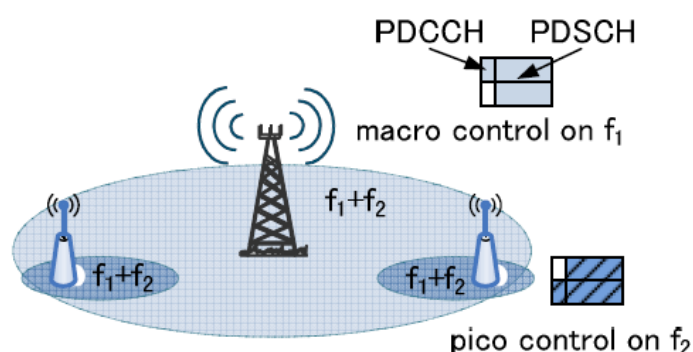


Figure 24. Deployment LTE-A HetNet using CA in a Co-Channel Deployment Scenario.

It is to be noted that some variations and alternatives may be found for the presented scheme. For example, the PDCCH may be transmitted at a reduced power or a hybrid solution of separating the PDCCH transmission for the different layers in the time domain in addition to the frequency domain may be in use.

6.6 CA AND MULTI-OPERATOR CORE NETWORK (MOCN)

The Multi-Operator Core Network (MOCN) was envisioned and included by 3GPP to facilitate network sharing between operators. Network sharing can help reduce operational cost by sharing infrastructure and spectrum and is becoming a popular solution for cost effectiveness or smooth transitions during mergers and acquisitions.

MOCN essentially allows operators to share the same radio resources while maintaining different core networks. An example is one e-NodeB and spectrum shared by two operators.

CA is aggregation of radio resources, and its functionality remains unchanged with MOCN. Capable devices may be scheduled for aggregated SCell resources irrespective of the operator's core network.

6.7 CA FOR ROAMING

CA's basic functionality remains unchanged in a roaming scenario. However, complexities lie in supporting multiple CA combinations on a single phone and radio design, additional mobility scenarios between these different band combinations and RF performance impacts of supporting different CA combinations. With these complexities to work through, global roaming with CA is still a few years out.

6.8 CA FOR OVERLAPPING BANDS

Certain LTE Bands as defined in Table 7 have overlapping components on UL or DL. An example is bands 12 and 17 where frequencies in Band 17 are also included in Band 12. This scenario requires a Multi Frequency Band Indicator (MFBI), which allows operation of devices in a multi-band environment.

In a multi-band environment, devices supporting MFBI are considered for CA as per their capabilities.

Table 7. CA for Overlapping Bands.

LTE Operating Band	Uplink (UL) operating band BS receive UE transmit		Downlink (DL) operating band BS transmit UE receive		Duplex Mode
	F _{UL_low} – F _{UL_high}		F _{DL_low} – F _{DL_high}		
12	699 MHz	– 716 MHz	729 MHz	– 746 MHz	FDD
17	704 MHz	– 716 MHz	734 MHz	– 746 MHz	FDD

7. CA DEVICE IMPLEMENTATIONS

7.1 UE CATEGORIES FOR CA

The 3GPP specification defines a number of UE categories for the purposes of creating device segments, while at the same time limiting fragmentation. In LTE Release 8 (the initial LTE release), five UE categories (Category 1-5) were defined. In Release 10, with the introduction of CA, up to 8x8 DL MIMO and up to 4x4 UL MIMO, three more categories (Category 6, 7 and 8) were added. More recently, in Release 11, two more categories (Category 9 and 10) were added.

The UE category defines a combined UL and DL capability. The DL parameters set by the UE category are given in Table 8, while the UL parameters set by the UE Category are given in Table 9. Both Table 8 and Table 9 are specified in 3GPP TS 36.306. [10]

Note that in order to preserve backward compatibility, a UE that indicates a Rel-10 UE Category must also indicate a Release 8 UE Category. A UE that indicates a Rel-11 UE Category must also indicate both a Release 10 and a Release 8 UE Category.

Table 8. Downlink Physical Layer Parameter Values Set by the UE Category.

UE Category	Maximum number of DL-SCH transport block bits received within a TTI	Maximum number of bits of a DL-SCH transport block received within a TTI	Maximum number of supported layers for spatial multiplexing in DL
Category 1	10296	10296	1
Category 2	51024	51024	2
Category 3	102048	75376	2
Category 4	150752	75376	2
Category 5	299552	149776	4
Category 6	301504	149776 (4 layers) 75376 (2 layers)	2 or 4
Category 7	301504	149776 (4 layers) 75376 (2 layers)	2 or 4
Category 8	2998560	299856	8
Category 9	452256	149776 (4 layers) 75376 (2 layers)	2 or 4
Category 10	452256	149776 (4 layers) 75376 (2 layers)	2 or 4

Table 9. Uplink Physical Layer Parameter Values Set by the UE Category.

UE Category	Maximum number of UL-SCH transport block bits transmitted within a TTI	Maximum number of bits of an UL-SCH transport block transmitted within a TTI	Support for 64QAM in UL
Category 1	5160	5160	No
Category 2	25456	25456	No
Category 3	51024	51024	No
Category 4	51024	51024	No
Category 5	75376	75376	Yes
Category 6	51024	51024	No
Category 7	102048	51024	No
Category 8	1497760	149776	Yes
Category 9	51024	51024	No
Category 10	102048	51024	No

Note also that the UE Category definition is common between FDD and TDD. The achievable DL peak data rate of a UE Category for FDD is 1,000 times the value given in the column titled “Maximum number

of DL-SCH transport block bits received within a TTI” in Table 8. The achievable UL peak data rate of a UE Category for FDD is 1,000 times the value given in the column titled “Maximum number of UL-SCH transport block bits transmitted within a TTI” in Table 9. For TDD, the achievable peak data rate of a UE Category further depends on the DL-UL configuration (DL/UL ratio).

Even though there is a strong association between UE Categories and the capability of aggregating a certain number of CCs, note that for most UE Categories, there is no standards-defined one-to-one linkage from UE Category to a number of aggregated CCs. For example, the standard does not preclude the possibility of a Category 1 UE supporting CA with five CCs, even though such a capability combination is highly unlikely in a product.

Table 10 provides information about the number of CC CA capability associated with each UE Category.

Table 10. Number of CC Aggregation Capability Associated with Each UE Category.

UE Category	Possible number of aggregated DL CCs	Typical number of aggregated DL CCs	Possible number of aggregated UL CCs	Typical number of aggregated UL CCs
Category 1	1...5	1	1...5	1
Category 2	1...5	1	1...5	1
Category 3	1...5	1, 2	1...5	1
Category 4	1...5	1, 2	1...5	1
Category 5	1...5	1	1...5	1
Category 6	1...5	2	1...5	1
Category 7	1...5	2	1...5	2
Category 8	5	5	5	5
Category 9	1...5	3	1...5	1
Category 10	1...5	3	1...5	2

It is expected that in 3GPP Rel-13, potentially also applicable to 3GPP Rel-12, new UE Categories will be added to target 600 Mbps peak data rate. That’s achievable with, for example, four CC DL CA with 2x2 MIMO, with three CC DL CA with 2x2 MIMO and 256-QAM or with two CC DL MIMO with 4x4 MIMO.

As it can be seen in Table 10, the number of DL and UL CCs need not be the same. The number of UL CCs configured for a UE must be less than or equal to the number of DL CCs.

For DL reception, the UE is required to have memory for storing soft channel bits for the purposes of HARQ combining of retransmissions. The size of the required memory is UE Category dependent. The memory size is not scaled up by the number of aggregated CCs but rather the same memory is divided up among the aggregated CCs with allocating equal memory size for each CC. There is no HARQ combining across different CCs participating in CA.

7.2 TRANSMITTER AND RECEIVER ARCHITECTURE FOR CA

For both the UE receiver and transmitter architecture, multiple options are possible depending on what part of the UE hardware is common for the processing of the CCs. As mentioned before, in general three different CA scenarios are considered:

1. Intra-band contiguous CA.
2. Intra-band non-contiguous CA.
3. Inter-band CA (non-contiguous).

7.2.1. RECEIVER

Table 11 describes two different receiver architectures options for the three scenarios.

Table 11. Possible UE Architecture for the Three Aggregation Scenarios.

Receiver Characteristics				
Option	Description (Rx architecture)	Intra-Band aggregation		Inter-Band aggregation
		Contiguous (CC)	Non-contiguous (CC)	Non-contiguous (CC)
A	Single (RF + FFT + baseband)	Yes	No	No
B	Multiple (RF + FFT + baseband)	Yes	Yes	Yes

Note that Option B is always applicable, while Option A is applicable only if the aggregated CCs have common timing, are adjacent in frequency and are on a common subcarrier raster. A common subcarrier raster necessitates that the carrier center frequency of the aggregated CCs is separated by a multiple of 300 kHz, which is the least common multiple of the LTE subcarrier spacing (15 kHz) and the LTE carrier raster (100 kHz). For Option B, no such restriction is necessary.

For intra-band contiguous aggregation, the UE may adopt either Option A: a single wideband-capable (e.g., >20MHz) RF front end (including mixer, AGC, ADC)⁹ and a single FFT, or alternatively Option B, i.e., multiple "legacy" RF front ends (20 MHz) and FFT engines. The choice between single and multiple transceivers depends on the comparison of power consumption, cost, size and flexibility to support other aggregation types.

For intra-band non-contiguous aggregation, using a single wideband-capable RF front end is undesirable due to the unknown nature of the signal between carriers. Another reason because multiple FFT would be necessitated by the possible timing offset between CCs.

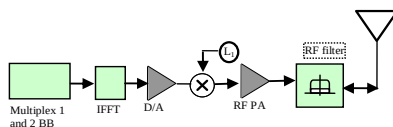
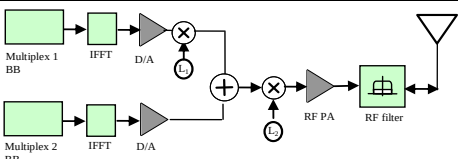
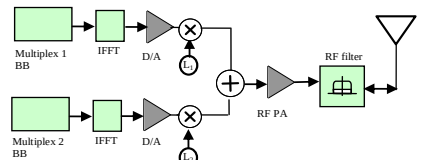
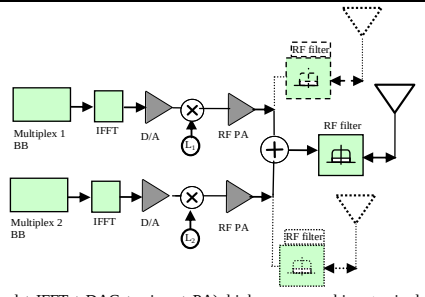
In the case inter-band CA, separate RF front ends are necessary.

⁹ AGC: Automatic Gain Control; ADC: Analog to Digital Convertor,

7.2.2. TRANSMITTER

Table 12 lists the various transmitter architectures options according to where the CCs are combined: at digital baseband, in analog waveforms before RF mixer, after mixer but before the Power Amplifier (PA) or after the PA. The examples show aggregation of two UL CCs.

Table 12. Possible UE Architectures in the Three Aggregation Scenarios [11].

Option	Description (Tx architecture)	Tx Characteristics		
		Intra Band aggregation		Inter Band aggregation
		Contiguous (CC)	Non contiguous (CC)	Non contiguous (CC)
A	 Single (baseband + IFFT + DAC + mixer + PA)	Yes		
B	 Multiple (baseband + IFFT + DAC), single (stage-1 IF mixer + combiner @ IF + stage-2 RF mixer + PA)	Yes	Yes	
C	 Multiple (baseband + IFFT + DAC + mixer), low-power combiner @ RF, and single PA	Yes	Yes	
D	 Multiple (baseband + IFFT + DAC + mixer + PA), high-power combiner to single antenna OR dual antenna	Yes	Yes	Yes + (depending on the specific EUTRA bands being aggregated),
X	OTHERS			

- Option A: The UE has one PA. Connected to the PA is a single RF chain (a zero-IF mixer, a wideband DAC, and a wideband IFFT¹⁰).
- Option B: The UE has one PA. The UE has two IFFTs, after which the UE combines analog baseband waveforms from the CCs (e.g., via a mixer operating at an IF¹¹ of roughly the bandwidth of the other component carrier). Then the resulting wideband signal is up-converted to RF.

¹⁰ IFFT: Inverse Fast Fourier Transform, computes the discrete inverse fast Fourier transform of a variable.

¹¹ IF: Intermediate Frequency

- Option C: The UE performs ZIF¹² up-conversion of each CC before combining and feeding into a single PA.
- Option D: The UE employs multiple RF chains and multiple PAs after which the high-power signals are combined and fed into a single antenna. PA coupling at the UE can be challenging for this option.

Note that other options are also possible. One example is combining the output of two IFFTs in the digital domain before the D/A conversion.

7.3 RF FRONT END IMPACTS

In a single carrier FDD (non-CA) scenario, a duplexer ensures that the transmission on the uplink does not interfere with the reception on the downlink. In a dual-band CA, say comprising of bands A and B, operators would need to ensure that the UL transmission on band A does not interfere with the DL reception on that band. Operators also need to ensure that it does not interfere with DL reception on band B.

Similar considerations apply to the UL transmission on band B. Even if one were to design a duplexer for each of the bands that ensures that neither DL band is affected, directly connecting two duplexers together can affect each other's filter characteristic, thereby losing the isolation that is needed to operate at reference sensitivity. Therefore, the general solution to this problem is a "quadplexer," except possibly in some cases with large frequency separation between the two bands, where a separate filter "diplexer" can be inserted between the antenna and the two individual band-specific duplexers. Such a quadplexer would need to be specially developed for each band combination and is likely to be significantly more complex and expensive compared to a duplexer.

We now consider a three-band case, such as three carriers on bands A, B and C. By the same arguments as for the dual-band case, a transmission on any band should not interfere with the reception on any other. Because the duplex filters are not tunable, the filtering solution has to be effectively designed to handle the case when all three bands are in use simultaneously. A "hexaplexer" would be necessary to support such tri-band communication. Similarly to the quadplexer case, when there is large frequency separation between at least one band and the other two bands, a separate filter "diplexer" can be inserted between the antenna and groups of individual duplexers. So for example, two duplexers and three band-specific duplexers may be used, or one diplexer and a combination of a quadplexer and a diplexer may be used.

Note that the above problem in the three-band case is not simplified even if we enforce a constraint that only two bands of the three bands are active at any given time. A UE that is designed to operate on band combinations A+B and A+C can't be automatically assumed to operate on band combination B+C. As a simple example, let band A be in the 1.9 GHz and let band B be lower 700 MHz and band C be upper 700 MHz. A UE could be designed with a filter to ensure sufficient isolation between the 700 MHz and 1.9 GHz carriers, but that filter cannot ensure isolation between the upper and lower 700MHz carriers. That is because the DL of the lower 700 MHz band (Band 12 or 17) and the DL of the upper 700 MHz band (Band 13) are adjacent to each other. Therefore, some signal splitting approach, as described below,

¹² Zero Intermediate Frequency (ZIF) up-conversion means no Intermediate stage of RF conversion between baseband and RF band. In the old day the baseband signal is modulated to IF signal first, then to RF signal before emitting in the Uplink. In the downlink, RF signal is demodulated to IF signal first then to baseband signal. All the mobile RF chipset do ZIF today.

needs to be implemented. Such signal splitting capability is not present in a non-CA switched RF solution. In addition, one low band antenna may not have enough bandwidth to cover low and lower band CA, and two low band antennas will be too big to fit in handset.

Obviously, considering the number of CA band combinations a UE may need to support, some simplification is necessary. A possible simplification is using a diplexer mentioned above. The diplexer is essentially a filter pair consisting of a low pass and a high pass filter that can separate bands that are sufficiently distant in frequency. For example, a diplexer can be used to separate a 700 MHz band from a 1.9 GHz band in the aforementioned example. Such a diplexer can be used for more than one CA band combination, making it can be a cost-effective solution. However, a significant drawback of using diplexers is the additional insertion loss, which can degrade both the receiver sensitivity and the transmitter output power by a noticeable degree. The degradation is present even in single carrier operation; therefore it might impact operation in legacy networks where CA might not even be configured. Overcoming such degradation in the UE may necessitate the redesign of the RF chains to compensate for the additional losses, which may negate the cost benefit of using a diplexer.

Even if the UL-to-DL isolation across band groups is solved by using a diplexer or by other means, the problem of splitting of DL signals or combining of UL signals within a band group still needs to be solved. Available methods for this are listed in Table 13.

Table 13. Signal Splitting and Combining Approaches.

Combiner / Splitter method	Guard band	Loss	Number of antennas	Notes
Direct combiner	Not needed	$10\log N$ dB	1	Difficult to realize for transmitter
Filter	Needed	Filter insertion loss	1	Not applicable to intra-band
Separate antenna	Not needed	Antenna coupling	N	Maximum isolation of ~10dB may be insufficient for transmitter Form factor, routing, and space constraints in a UE may not suit a multiple antenna solution

7.4 BATTERY IMPACTS

CA operation has an impact on UE current consumption due to more receiver activity, more processing and potentially more UL activity. The impact varies based on the receiver and/or transmitter architectures. The current consumption can, in some CA configurations, approach the power consumption obtained by multiplying the non-CA current consumption by the number of aggregated CCs.

The power consumption is influenced by the CA configuration settings, as well as the state of the CC. Configuring the UE with CA when there are small volumes of data traffic for a UE should always be done with care using CA configurations settings aimed to avoid large UE power consumption impact.

A mechanism introduced to reduce the impact on the UE battery life is MAC activation/deactivation of a CC while the UE is configured with CA. Each CC (secondary cells only) can be individually activated or deactivated. When the CC is activated the activity (and thereby UE power consumption) follows that of

the primary cell while when the CC is deactivated the activity is reduced, thereby reducing the UE power consumption impact. When the UE is configured with CA, any newly added CC is always initiated in a deactivated state. In order to start scheduling the UE on that CC, it is necessary to first activate the CC by performing MAC activation. The delay between the activation command and the availability of a CC is at maximum 24-34 ms. The UE may likely be ready prior to the MAC activation delay. The delay of deactivating a CC is much shorter, and up to 8 ms.

When a CC is deactivated, the UE is still required to perform RRM measurements on the carrier. That means – as for any other RRM measurements – that the UE needs to periodically turn on the RF chain to measure a deactivated carrier. In a UE using a single RF chip, the events of turning on/off a deactivated carrier may create interruptions to the other CCs (including the primary CC). If the UE were to completely avoid these interruptions, it can only turn off the RF chain of a deactivated carrier according to the primary CC activity, thereby reducing the battery-saving benefits of MAC activation/deactivation. The impact depends on the CA configuration settings used in the network and UE power savings can be enabled also for single RF chip solutions. A UE not using the single RF chip solution will not face similar problems of creating interrupts and would be able to get full benefits of applied MAC activation/deactivation.

Note that in the case of UL CA, there is further impact on battery life when the PA output has to be increased to overcome diplexer insertion losses and/or when the individual PAs have to operate in a less power efficient regime than a single PA would have to for the same total output power.

7.5 LEGACY DEVICE IMPACTS

In general, there is little impact on legacy devices, as long as the guard band between any carriers conforms to the Release 8 specification and the CA configuration for each legacy UE is in conformance with the UE's indicated capabilities. The latter is usually possible because the CA configuration is always UE-specific. For example, if a set of four CCs is available in a network, some UEs can be configured with no CA, others with two-CC CA, still others with three-CC CA and so on.

Note that as part of the UE CA capabilities, the UE can indicate, for example, that it can aggregate Band A and Band B in a 2 CC DL + 1 CC UL configuration, where the UE supports Band A DL together with Band B DL when the UL is in Band A but it does not support the same when the UL is in Band B. The eNB needs to use CA configuration that conforms to these capabilities.

Changing the guard band between component carriers, as mentioned in Section 4, impacts legacy UEs that may attempt to operate on either component carrier. This is true irrespective of whether CA is configured for the legacy UE or whether the UE even supports CA.

In order to enable early deployment of CA-capable UEs, certain reduction of CA bandwidth combinations were allowed in Release 10. For example, a UE that can operate on a 20 MHz carrier without CA may be able to operate only on 10MHz+10MHz with CA. When a candidate CC is converted to 20 MHz in the future, these UEs will no longer be able to operate in CA mode. In this particular example, the peak throughput of the UE would not be reduced but in other cases it might be.

8. CA FUTURE ENHANCEMENTS

8.1 CA ENHANCEMENT BEYOND FIVE CARRIERS

CA work started in Rel-10 with the basic CA feature support, enabling aggregation of up to five carriers of the same frame structure. Extensions of the basic CA framework currently enable inter-band TDD CA with different UL-DL configurations, CA with multiple UL timing advance (in Rel-11) and aggregation of carriers with different frame structures through FDD-TDD CA (in Rel-12).

Also, 3GPP RAN WG 4 has been progressing well lately with the CA requirements specification (especially in DL direction) with several 3-DL CA work items being completed, as well as several 4-DL CA configurations are under discussion. From this respect, the RAN1 and RAN2 specifications for LTE CA start soon being a limiting factor for further LTE CA configurations, especially for DL.

Licensed-Assisted Access (LAA) has created a lot of interest in extending the LTE CA feature toward capturing the spectrum opportunities of unlicensed spectrum in the 5 GHz band. WLANs operating in the 5 GHz band currently support 80 MHz in the field, with 160 MHz to follow in Wave 2 deployment of IEEE 802.11ac.

There are also other frequency bands, such as 3.5 GHz [12] [13], where aggregation of more than one carrier on the same band is possible, in addition to the bands already widely in use for LTE.

Enabling the utilization of at least similar bandwidths for LTE in combination with LAA through CA calls for extending the CA framework to support more than five (at least DL) carriers. The extension of the CA framework beyond five carriers would give operators an opportunity to more efficiently utilize the available spectrum for communication needs.

Not all the CA aspects scale directly with an increasing number of component carriers. Besides the extension beyond five carriers, the studies should investigate potentially new ways on how to efficiently manage and operate an increased number of component carriers. Such studies may include extension of DL and UL control signalling as well as HARQ feedback, beside others. Also, Layer 2/layer 3 aspects regarding efficient CC management can be studied. A Study Item is being proposed at the September 2014 RAN Plenary meeting. [14][15]

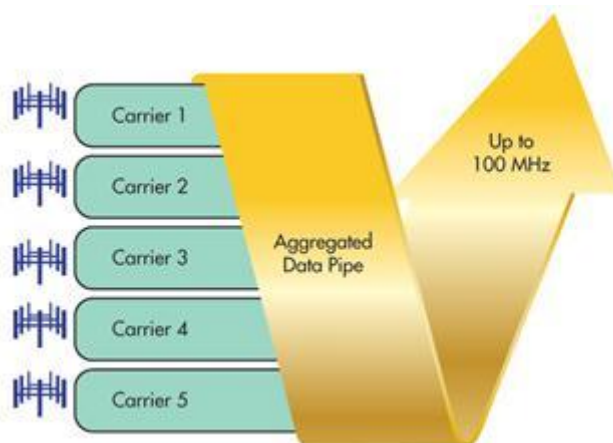


Figure 25. Five-Carrier CA.

8.2 CA WITH LICENSED ASSISTED ACCESS

The main goal for LTE in unlicensed spectrum is to leverage the relatively large amount of unlicensed spectrum in order to provide better mobile service. Given the transmit power restrictions imposed on unlicensed bands, it is expected to be primarily deployed in small cells. No standards exist for LTE in unlicensed spectrum and the various technical and regulatory items need to be addressed in the future. However, the mobile industry is discussing certain technical proposals which include an LTE carrier in unlicensed spectrum (as a secondary LTE cell) combined with an LTE carrier in licensed spectrum (as the primary LTE cell) using CA, with all control channels on the licensed spectrum LTE carrier.

The LTE carrier on the unlicensed spectrum can either be a Supplemental Downlink (SDL) or can be used for DL and UL transmissions. Due to the tight combination of licensed and unlicensed carriers and due to control channels being on the licensed primary cell, the technology is also called Licensed Assisted Access. Specifically, industry proposals do not describe “stand-alone” LTE in an unlicensed spectrum system where there is no LTE carrier. A key technical requirement and benefit targeted by these proposals is to extend the management, QoS mechanisms¹³, security and mobility aspects of LTE into the unlicensed spectrum. These benefits are the key reasons brought forth by proponents.

The primary targeted unlicensed band is the 5 GHz band. However, the regulations for usage of this band vary across the world and standards will have to account for them. The industry proposals describe the need for Wi-Fi co-existence and articulate how the two unlicensed technologies may complement each other. It cannot be overstated that LTE in unlicensed spectrum can only be a supplementary technology to LTE, which today is operable only in licensed spectrum. Specifically, development (and deployment) of LTE in unlicensed spectrum does not decrease the need for additional licensed spectrum because high quality services with high predictability can only be provided via the usage of licensed spectrum.

Most industry proposals describe using LTE in a CA or SDL implementation in which LTE in the unlicensed band operates in conjunction with LTE in the licensed band. There have been some industry proposals for a “stand-alone” mode of operation, in which LTE operates in unlicensed spectrum without the need for LTE in licensed spectrum.

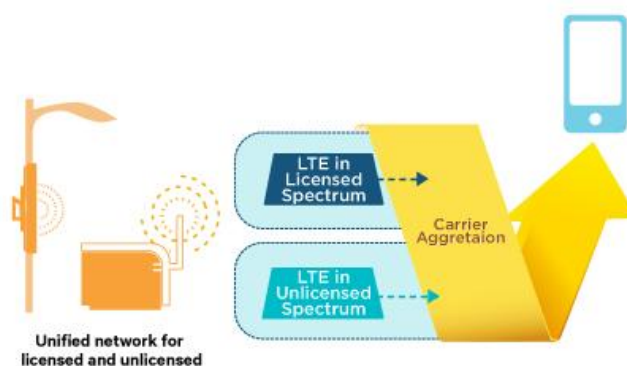


Figure 26. Unified LTE Network for both Licensed and Unlicensed Spectrum [16].

¹³ It should be noted that services requiring QoS levels other than best efforts would typically not be transmitted over unlicensed spectrum due to the lack of predictability

8.3 DUAL CONNECTIVITY [3]

In LTE, dual connectivity, also known as Inter-site CA, is being introduced in Rel-12 to achieve CA between sites. This is an attractive solution for Heterogeneous Networks (HetNets) with no ideal backhaul network. Dual connectivity allows mobility management to be maintained on the macro layer while aggregating small cells to provide extra user plane capacity, increasing the throughput. The concept optimizes performance by combining the benefits of macro cell coverage and small cell capacity. Based on increasing the bandwidth through CA, inter-site CA can provide a cell edge gain of 50 percent even in loaded networks.



Figure 27. Simultaneous Connection to the Macro and Pico layers.

Figure 28 depicts how Dual Connectivity is achieved. The radio protocols of the user plane are split between the Master eNB (MeNB), typically a macro cell, and the Secondary eNB (SeNB), typically a small cell. This gives more flexibility to radio bearers carrying user data. They can either use resources of the macro cell only (depicted in grey), of the small cell only (depicted in cyan) or aggregate both (depicted in blue), depending on whether coverage, offload or throughput is to be favored.

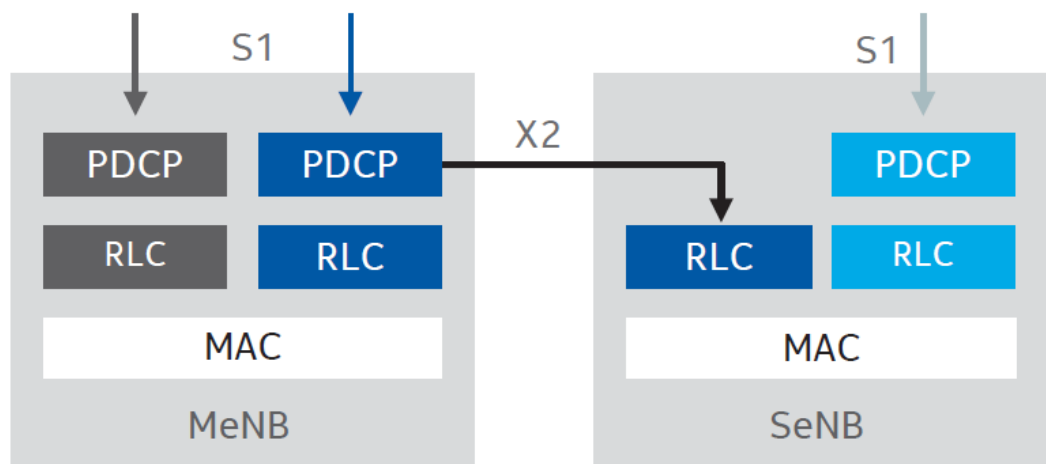


Figure 28. Dual Connectivity.

8.4 CA FOR 5G

The 5th generation cellular technologies (5G) have not yet been defined, but it is likely that CA will play an important role in these technologies, too. Similar to CA in LTE-Advanced, 5G CA should also enable using larger parts of the spectrum by a single device.

Because the range of candidate frequencies for aggregation may become much wider in 5G, it is possible that the aggregated carriers would use different signal formats to better adapt to the operating frequency bands. When aggregating carriers with different signal formats, the CA framework may need to become more flexible compared to LTE-Advanced.

As discussed in Section 8.3, in LTE, dual-connectivity is being introduced in Rel-12, which already allows more independent operation on different carriers. In dual-connectivity, unlike in CA, the physical layer and MAC layer signaling is not shared across certain groups of component carriers. For example, the channel state information describing the DL channel within one carrier group has to be sent on a carrier within the same carrier group. This way of operation reduces the need for tight, fast time scale coordination between carrier groups. Note the dual-connectivity operation defined in Release 12 requires at least one UL component carrier per carrier group, while in CA, a single UL among all component carriers is sufficient. The same dual-connectivity framework might be evolved to allow aggregation of frequencies with large separation.

Similarly, future cellular technologies may provide more options to aggregate disparate air-interfaces. In these cases also, a more flexible framework is needed to enable simultaneous operation with different signal formats.

Another future use case for CA is to overcome the unreliability of individual links. This might be necessitated by using frequencies with adverse propagation conditions or adverse interference conditions, the latter due to unlicensed use, for example. Even with reliable individual links, CA may be needed to further improve radio channel availability and QoS for mission-critical applications.

An important potential future enhancement for CA is to mitigate the negative impact of CA on device power consumption. This should include both advances in RF technology and improvements in the physical layer/MAC procedures. The improved procedures should enable faster turning on/off of a given CC. Further improvements may also allow the UE to determine the best CA operating configuration, especially in cases where there is only loose coordination between the aggregated eNBs. Therefore the eNBs may lack information about other links the UE is operating on.

9. CONCLUSION AND RECOMMENDATIONS

The larger bandwidth provided by LTE-Advanced CA obviously results in improved user data rates. But equally important, CA is a powerful feature that enables more flexible and optimal utilization of frequency assets. Especially, non-contiguous CA offers new opportunities to use more and more frequency resources for LTE in different bands.

CA for LTE-Advanced is fully backward compatible, which essentially means that legacy Rel-8 terminals and LTE-Advanced terminals can co-exist. The latter is achieved by relying on MAC level CA with independent Release 8 compliant HARQ and link adaptation per CC. This also implies that CA is transparent from layer 3 and up for the user plane. A flexible layered approach for managing the CCs per LTE-Advanced user is defined, offering configuration of CCs per user via RRC signaling, followed by MAC signaling for activation/de-activation of CCs configured as SCells. The aforementioned control

procedures facilitate efficient power management of terminals, so they are not always mandated to operate at their full bandwidth capability.

CA is clearly a technology that will be a solution to the fragmented spectrum operators today own, enabling larger spectrum pipes and more efficient network infrastructures. This is evident from the great progress in the development of this technology throughout all facets of the industry from the standards to network and device infrastructures and quickly gaining momentum, with operators around the world deploying the technology.

CA has also been designed to be a future-proof technology with great potential into and beyond Release 12. CA will extend to multiple carriers, aggregation of varied licensed spectrum (possibly unlicensed spectrum) and will be the technology to enable IMT-Advanced speeds of 1Gbps. Looking forward, there will be multiple and varied deployments of CA tailored to operators' specific requirements. CA will also be a critical component as new bands are defined and the search for new spectrum resources grows with upcoming spectrum auctions and frequent spectrum swaps between operators.

However, this seemingly simple concept will continue to stretch the realm of network and device implementations. Strong collaboration across the industry for further innovations, evolutions and optimization will drive this as a main stream technology across devices, infrastructure and operators while bringing consumers network speeds and efficiency that will enable superior applications and further innovation and evolution of the smartphone. In particular, operators are advised to carry out consultations with their RAN infrastructure vendors, as well as their device and chipset manufacturers, when developing their CA implementation plans.

More specific areas for collaboration include further evolution of the standards to develop aggregation of multiple carriers, extending into the unlicensed bands and aggregation with HetNet scenarios. On the device and chipset front, new innovations are essential in the radio front-end technologies to enable CA on multiple bands without loss of performance. Device power consumption is another potential future enhancement. Infrastructure and operators can continue to optimize scheduler and mobility between different aggregated bands as best tailored to their network and more innovations on device and infrastructure in selecting the best CA combinations.

In the coming years, mobile data usage will continue to grow exponentially, new applications and services requiring higher speeds will continue to evolve and mobile consumers across the world grow. Developing flexible and forward-looking technologies such as Carrier Aggregation today will lead the path to LTE-Advanced speeds, 5G and beyond.

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11. GLOSSARY

The following acronyms and technical terms are used in this paper.

- Carrier Aggregation (CA): Technology to enable spectrum expansion by aggregation of spectrum across multiple bands.
- ATBC (Aggregated Transmission Bandwidth Configuration): is the total number of aggregated Physical Resource Blocks.
- LAA (Licensed-Assisted Access): Technology for expanding spectrum by use of LTE in unlicensed spectrum either by Supplemental Downlink (SDL) or for Downlink (DL) and Uplink (UL) transmissions
- Supplemental Downlink (SDL): Technology to enhance downlink capability of mobile broadband networks by use of unpaired downlink spectrum.
- P-Cell: Primary Cell, refers to the serving cell in a carrier aggregation scenario
- SCell: Secondary Cell refers to the cell adding additional spectrum in a carrier aggregation scenario
- CC (Component Carrier): Each aggregated carrier is also referred to as a component carrier
- PCC: Primary Component Carrier
- DL-PCC (Downlink Primary Component Carrier): Downlink Carrier corresponding to PCell
- UL-PCC (Uplink Primary Component Carrier): Uplink Carrier corresponding to PCell
- Heterogeneous Networks (HetNet): A radio access network that consists of layers of different-sized cells ranging from macrocells to picocells and femtocells.
- ICIC (Inter-cell interference coordination): 3GPP feature to improve cell edge performance and mitigate interference between macro and small cells in a HetNet scenario.
- MOCN (Multi-Operator Core Network): 3GPP Feature that allows multiple operators to share the same spectrum resources while maintaining separate core networks.
- MFBI (Multi-Frequency Band Indicator): 3GPP feature to allow operation of LTE networks and devices in a multi-band environment
- MIMO (Multiple Input Multiple Output): Radio technology that involves use of multiple antennas at both the network and device to improve communication performance and achieve higher speeds.
- AGC (Automatic Gain Control): Component close to antenna in the receive path which controls and adjusts the signal strength of the received signal before input to the next stage.
- ADC (Analog to Digital Converter): Performs analog to digital conversion of signal between RF and baseband in the downlink.
- PDCCH (Physical Downlink Control Channel): Control Channel carries the downlink and uplink allocation information for the terminal.

- PDSCH (Physical Downlink Shared Channel): Shared data channel allocated for users on a dynamic basis.
- TTI (Transmission Time Interval): Duration of transmission on a radio link, 1 ms for LTE networks
- QoS (Quality of Service): Framework to define and guarantee performance measures in an LTE network.
- SRS refers to Sounding Reference Signals which are transmitted on the uplink containing a known sequence to both network and devices and allow the network to estimate the quality of the channel at different frequencies. SRS, may be transmitted periodically in a wider bandwidth (beyond the UL resources blocks allocated for UL data transmission) and also when there is no UL data transmission, so the channel information obtained from SRS is a good input to the uplink scheduler at the eNB.
- CQI is Channel Quality Indicator. PMI stands for Pre-coding Matrix Indicator and RI is Rank Indicator. These three values are computed on the fly in LTE systems and used to try to optimize resource allocation among the various user end devices (UEs) that are requesting service. The CQI is reported by UE to eNB. UE indicates modulation scheme and coding scheme to eNB, that would allow it to demodulate and decode the transmitted downlink data with maximum block error rate 10%. The CQI reporting can be based on PMI and RI. For PMI, the UE indicates to eNB which precoding matrix from a predefined codebook should be used for downlink transmission by the eNB which is determined by RI. For RI the UE indicates to eNB, the number of spatial layers that should be used for downlink transmission to the UE.
- PTI stands for Precoding Type Indicator and it is a new feedback indicator added for LTE-Advanced, which is reported by the UE and allows distinguish slow from fast fading environments.
- UL-SCH is a MAC uplink transport channel and is carried over the Physical Uplink Shared Channel. The UL-SCH is used for UL transmission of UE according to scheduling guidelines given by the eNB.
- RACH (Random Access Channel): This channel is used for the execution of the Random Access Procedure in which the network (eNB) for the first time knows that some UE is trying to get access.
- IF: Intermediate Frequency
- Zero Intermediate Frequency (ZIF) up-conversion means no Intermediate stage of RF conversion between baseband and RF band. In the old day the baseband signal is modulated to IF signal first, then to RF signal before emitting in the Uplink. In the downlink, RF signal is demodulated to IF signal first then to baseband signal. All the mobile RF chipset do ZIF today.
- IFFT (Inverse Fast Fourier Transform): computes the discrete inverse fast Fourier transform of a variable.

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